Contents

1 Normed spaces

1.1 Review of norms

The vector spaces we study here will be over $\mathbb{R}$ or $\mathbb{C}$ but it doesn't really matter which, but they will often be infinite dimensional.

Recall the following definition of a norm: A norm is a function from a vector space $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ satisfiny

i) $\|x\| = 0 \iff x = 0$

ii) $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{R}$ or $\lambda \in \mathbb{C}$ depending on which field we are working in.

iii) $\|x + y\| \leq \|x\| + \|y\|$ i.e. the norm is "subadditive" or satisfies the triangle in equality.

We can think of this as length, but since we will be working in infinite dimensional spaces it is a bad idea to rely on this intuition too much.

The following are examples of vector spaces we might look at in this course and possible norms.

Example. Let S be any set, then the vector space of real valued functions $f : S \to \mathbb{R}$ has a norm given by $\|f\|_\infty = \sup_{s\in S} |f(s)|$

Example. $C[0, 1]$, the set of all continuous functions on $[0, 1]$ is a vector space and a possible norm is the absolute value of its maximum (maximum and not supremum by the extreme value theorem). In fact we can assign any norm for

$1 \le p < \infty$ by $\|f\|_p = \left(\int_0^1 |f(t)|^p\right)^{1/p}$

Example. $C^k[0,1]$, the space of k times continuously differentiable functions on $[0,1]$ can have norm given by

$$\sup_{t\in[0,1]} \left(|f(t)| + |f'(t)| + ... + \left|f^{(k)}(t)\right|\right)$$

Example. The set of Riemann integrable functions on $[0,1]$ is a vector space but the function $f \mapsto \int |f|$ is not a norm since it can be 0 without f being 0. However, doing a trick like we did in the probability and measure course where we define an equivalence relation $f \sim g$ if $\int |f-g| = 0$ and take the equivalence classes gives us an actual normed space.

Example. For $1 \le p < \infty$ the set of sequences $l_p = \{(x_1, x_2, ...) : \sum_{i=1}^{\infty} |x_i|^p < \infty\}$ has a norm given by $\left(\sum_{i=1}^{\infty} |x_i|^p\right)^{1/p}$ and this is indeed a norm by Minkowski's inequality from the Probability and measure course - since the sum is effectively an integral with respect to the counting measure. In the case $p = 2$ this gives us an inner product and therefore a hilbert space. We can define $l_\infty = \{(x_1, x_2, ...) : \sup |x_i| < \infty\}$ i.e. the vector space of all bounded sequences.

Proposition. (Weighted AM-GM inequality) Let $a_1, a_2, ..., a_n > 0$ and let $\lambda_1, \lambda_2, ..., \lambda_n \in [0,1]$ satisfy $\lambda_1+\lambda_2+...+\lambda_n = 1$, then we have

$$a_1^{\lambda_1} a_2^{\lambda_2} ... a_n^{\lambda_n} \le \lambda_1 a_1 + \lambda_2 a_2 + ... + \lambda_n a_n$$

Proof. We put log of both sides to get that the inequality we aim to prove is equivalent to

$$\lambda_1 \log(a_1) + \lambda_2 \log(a_2) + ... + \lambda_n \log(a_n) \le \log(\lambda_1 a_1 + \lambda_2 a_2 + ... + \lambda_n a_n)$$

Since log is a concave function (this just means $-\log$ is a convex function), Jensen's inequality implies the result.

□

Recall the following inequalities and definitions from the Probability and Measure and Analysis II course, applied to the case of the integral with respect to the counting measure on finite sequences

Proposition. (Holder's inequality) Let a_i, b_i be sequences of complex numbers and let p, q be such that $\frac{1}{p} + \frac{1}{q} = 1$ and $1 < p < \infty$. Then

$$\sum_{i=1}^{n} |a_i b_i| \le \left(\sum_{i=1}^{n} |a_i|^p\right)^{\frac{1}{p}} \left(\sum_{i=1}^{n} |b_i|^q\right)^{\frac{1}{q}}$$

Proposition. (Minkowski's inequality) Let the same conditions as above be true, then

$$\left(\sum_{i=1}^{n} |a_i + b_i|^p\right)^{1/p} \le \left(\sum_{i=1}^{n} |a_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |b_i|^p\right)^{1/p}$$

Definition. V is a Banach space if it is complete with respect to its norm i.e. Cauchy sequences converge. It is a Hilbert space if it also has an inner product structure.

Definition. Let $\|\cdot\|$ and $\|\cdot\|'$ be norms, then they are said to be equivalent if there are $c_1, c_2 > 0$ such that for all x we have $c_1 \|x\| \le \|x\|' \le c_2 \|x\|$. It is easy to check that this is an equivalence relation.

Note that if X and Y are complete metric spaces then so is $X \times Y$ with the metric given by

$$d((x_1, y_1), (x_2, y_2)) = d(x_1, x_2) + d(y_1, y_2)$$

since if (x_n) is Cauchy in X and (y_n) is Cauchy in Y then eventually the terms in both get within $\varepsilon/2$ of eachother so (x_n, y_n) is Cauchy in $X \times Y$ so if the limits are x, y repspectively then (x, y) is in $X \times Y$ so $X \times Y$ is complete.

Proposition. Any 2 norms on $\mathbb{R}^n$ or $\mathbb{C}^n$ are equivalent.

Proof. It suffices to show that a given norm $\|\cdot\|$ is equivalent to the l_1-norm $\|\cdot\|_1$ i.e. the sum of absolute values of the entries.

Suppose v, w are such that $\|v-w\|_1 \leq \delta$, then given an arbitrary norm $\|\cdot\|$ we have

$$\|v-w\| = \left\|\sum (v_i - w_i) e_i\right\| \leq \sum |v_i - w_i| \|e_i\| \leq \max_i \|e_i\| \sum |v_i - w_i|$$

This proves that there is an 'upper' constant c_2. In fact, it proves that $x \mapsto \|x\|$ is continuous with respect to the l_1-norm. Therefore, on the unit sphere[1], i.e. the set $\{x : \|x\|_1 = 1\}$, the function $x \mapsto \|x\|$ is bounded and attains its bounds, and its lower bound cannot be 0 by definition of a norm, and so its minimum is some non-zero constant and this works as our c_1. This implies that if $\|x\|_1 = 1$ then there are $c_1, c_2 > 0$ with $c_1 \|v\|_1 \leq \|v\| \leq c_2 \|v\|_1$ (by dividing v by $\|v\|_1$ and applying the result for the unit "sphere")

$\square$

Corollary. Any finite-dimensional normed space is a Banach space (since we showed in the Linear Algebra course that they must be isomorphic to $\mathbb{R}^n$ or $\mathbb{C}^n$ and clearly equivalent norms preserves whether a sequence is Cauchy)

1.2 Norms and convex bodies

Theorem. There is a natural correspondence between norms in $\mathbb{R}^n$ and closed, bounded, centrally symmetric (meaning $x \in K \iff -x \in K$) convex bodies containing an open ball about the origin.

Proof. The set $\{x : \|x\| \leq 1\}$ is convex because if x, y satisfy it and $t \in [0,1]$ then

$$\|tx + (1-t)y\| \leq t\|x\| + (1-t)\|y\| \leq 1$$

On the other hand, given such a convex body K, we define $\|x\| = \min_\lambda \{x \in \lambda K\}$. Since the interior is non-empty and the set is closed this is well defined. By convexity, we have $\lambda K + \mu K$ is a subset of $(\lambda + \mu) K$ for $\lambda, \mu \geq 0$ - to see this note that an arbitrary point $\lambda K + \mu K$ is for $x, y \in K$ equal to the point $\lambda + \mu \left(\frac{\lambda x + \mu y}{\lambda + \mu}\right) \in (\lambda + \mu) K$. This establishes the triangle inequality so it is a norm.

$\square$

1.3 Examples of Banach spaces

Example. Continuous functions with the infinity norm are a Banach space because the uniform limit of continuous functions is continuous. However, with the $\|\cdot\|_1$ norm it is not a Banach space because of the sequence $f_n(x) = n \int_0^x 1_{(0,1/n)}(t)dt$ is Cauchy its limit is the Heaviside step function which (contrary to what they want you to believe in some courses) is not continuous.

In fact, on $C[0,1]$ the $\|\cdot\|_1$ and $\|\cdot\|_\infty$ norms are not equivalent, for if they were the above example would be a Cauchy sequence in one but not the other.

Proposition. l^p is complete

Proof. The $p = \infty$ case is easy by the triangle inequality. Otherwise, let (x_n) be a Cauchy sequence in l^p. Then since each x_n is a sequence write them as $x_i = (x_{i,1}, x_{i,2}, ...)$. We must prove that there exists $x^* \in l^p$ such that $\|x_n - x^*\|_p \to 0$.

Note that for any fixed i, $|x_{n,i} - x_{m,i}| \leq \|x_n - x_m\|_p$ and therefore $(x_{n,i})_{n=1}^\infty$ is a Cauchy sequence. Since $\mathbb{R}$ and $\mathbb{C}$ are complete, there exists x_i^* such that $x_{n,i} \to x_i^*$. So we define $x^* = (x_1^*, x_2^*, ...)$ and we claim that $x_n \to x^*$ in l^p.

[1] Well, technically not actually a sphere, that would be if this was the l^2-norm

To do this, let m be sufficiently large such that $\|x_m - x_n\|_p \leq \varepsilon$ for all $n \geq m$ and let R be arbitrary. Then

$$\sum_{i=1}^{R} |x_{m,i} - x_{n,i}|^p \leq \varepsilon^p$$

Let $n \to \infty$ then we get

$$\sum_{i=1}^{R} |x_{m,i} - x_i^*|^p \leq \varepsilon^p$$

Now let $R \to \infty$ then we get

$$\sum_{i=1}^{\infty} |x_{m,i} - x_i^*|^p \leq \varepsilon^p$$

Therefore for m sufficiently large we have

$$\|x_m - x^*\|_p \leq \varepsilon$$

□

Note that this implies that l^2 is actually a Hilbert space since it is complete.

1.4 Completion of normed spaces

Let $(V, \|\cdot\|)$ be a normed space, then we define the completion of V as follows. Let $\mathcal{C}_V$ be the set of Cauchy sequences in V. Then this is a vector space. We define an equivalence relation of sequences where $(x_n) \sim (y_n)$ if $(x_n - y_n) \to 0$ and define a vector space $\overline{V}$ as the set of equivalence classes.

We define a norm on $\overline{V}$: Given x an element in $\overline{V}$ let (x_n) be any Cauchy sequence in that equivalence class and define $p(x) = \lim_{n\to\infty} \|x_n\|$ which is a limit that exists as (x_n) is a Cauchy sequence, then $p(x)$ is the norm of x in $\overline{V}$, and it doesn't depend on which element of the equivalence class you are in since if $x \sim y$ then $\|x_n\| - \|y_n\| \to 0$ so their limits are equal, and it is easy to see that p satisfies the following properties which make it actually a norm:

i) $p(x) = 0 \iff (x_n \sim 0)$

ii) $p(\lambda x) = |\lambda| p(x)$

iii) $p(x+y) \leq p(x) + p(y)$

This means $\overline{V}$ is a normed space. We will check soon that it is actually complete and has the properties we want.

Although this isn't a perfect analogy since before constructing the reals we can't construct a norm, it's like defining the real numbers by Cauchy sequences of rational numbers and their equivalence classes. Essentially, we are trying to fill in gaps, as made precise by the following definition.

Define a map $\phi : V \to \overline{V}$ sending $x \to (x, x, x, x, x, ...)$

Proposition. The image of $\phi(V)$ is dense in $\overline{V}$

Proof. Let $[x] \in \overline{V}$ and $\varepsilon > 0$. Then x corresponds to a Cauchy sequence so that there is an n_0 such that $m, n \geq n_0$ implies $\|x_n - x_m\| < \varepsilon$. Then

$$p(\phi(x_{n_0}) - [x]) - p([(x_{n_0} - x_i, x_{n_0} - x_2, x_{n_0} - x_3, ...)]) \leq \varepsilon$$

Thus we can find an element of $\phi(V)$ in any open neighbourhood of x so the image is indeed dense.

□

Proposition. $(\overline{V}, p)$ is complete

Proof. Let $([x^n])$ be Cauchy in $\overline{V}$, then (x_n) is Cauchy in V for all n. Since $\overline{\phi(V)} = \overline{V}$, for each $n \in \mathbb{N}$ there exists $z_n \in V$ such that $p(\phi(z_n) - [x^n]) \leq \frac{1}{2^n}$.

Then we have

$$\begin{aligned} \|z_n - z_m\| &= p(\phi(z_n) - \phi(z_m)) \\ &= p(\phi(z_n) - [x^n] + [x^n] - [x^m] + [x^m] - \phi(z_m)) \\ &\leq p(\phi(z_n) - [x^n]) + p([x^n] - [x^m]) + p([x^m] - \phi(z_m)) \end{aligned}$$

This tends to 0 so (z_n) is Cauchy in V. Therefore it corresponds to an element $[z]$ in $\overline{V}$. To complete the proof we will show that this is actually a limit of x^n so that $p([z] - [x^n]) \to 0$. We have

$$p([z] - [x^n]) = \lim_{m\to\infty} \|z_m - x_m^n\|$$

$$\leq \limsup_{m\to\infty} \|z_m - z_n\| + \limsup_{m\to\infty} \|z_n - x_m^n\|$$

Both terms tend to 0 since z is Cauchy and we defined it so that the second term would also go to 0, so done.

□

Definition. Let V be any normed space, then a completion of V is a triple $(\overline{V}, p, \phi)$ such that $\phi : V \to \overline{V}$ is linear, preserves distances and its image is dense in V, and $(\overline{V}, p)$ is complete.

Definition. An isometric isomorphism is an isomorphism between Vector spaces that preserves the norm

Theorem. The completion of a normed space V is unique up to an isometric isomorphism

Proof. Let $(\overline{V}_1, p_1, \phi_1)$ and $(\overline{V}_2, p_2, \phi_2)$ be two such completions. We want to find an isometric isomorphism between these two spaces.

Define a map $\psi_0 : \phi_1(V) \to \phi_2(V)$ be sending $\psi_0(\phi_1(x)) = \phi_2(x)$. Since ϕ_1 is distance-preserving it is injective meaning this map is well defined and bijective between the dense subspaces $\phi_1(V)$ and $\phi_2(V)$. Furthermore, since ϕ_1 and ϕ_2 preserve the original norm of V, ψ_0 is an isometry.

We define this mapping ψ_0 to a map $\psi : \overline{V}_1 \to \overline{V}_2$. Given $y \in \overline{V}_1$, there exists a sequence (x_n) in V such that $\phi_1(x_n) \to y$ in $\overline{V}_1$. Since the sequence $\phi_1(x_n)$ converges, it must be a Cauchy sequence in $\overline{V}_1$. Also, $\psi_0(\phi_1(x_n)) = \phi_2(x_n)$ is Cauchy in $\overline{V}_2$ and therefore must converge to some limit $z \in \overline{V}_2$. We will define our extended map $\psi : \overline{V}_1 \to \overline{V}_2$ by sending each y to its corresponding z.

To prove that this extended map is well defined, we must prove that it does not depend on the Cauchy sequence. This follows from the fact that if x^1, x^2 tend to the same limit then $x_n^1 - x_n^2 \to 0$ so $p_1(\phi_1(x_n^1 - x_n^2)) \to 0$ as $n \to \infty$. Since ψ_0 is an isometry on the dense subspace,

$$p_1(\phi_1(x_n^1) - \phi_1(x_n^2)) = p_2(\phi_2(x_n^1) - \phi_2(x_n^2)) \to 0$$

Therefore the sequences $\phi_2(x_n^1)$ and $\phi_2(x_n^2)$ converge to the same limit in $\overline{V}_2$ so ψ is well defined.

To prove ψ is an isometric isomorphism note that it is linearity since limits are linear and ϕ_1 and ϕ_2 are linear. It is isometric because

$$p_2(\psi(y)) = p_2\left(\lim_{n\to\infty} \phi_2(x_n)\right) = \lim_{n\to\infty} p_2(\phi_2(x_n))$$

The last equality is because norm is a continuous function. Since ϕ_1 and ϕ_2 preserve distances we have

$$= \lim_{n\to\infty} p_1(\phi_1(x_n)) = p_1\left(\lim_{n\to\infty} \phi_1(x_n)\right) = p_1(y)$$

Therefore ψ is an isometry. It remains to check that it is actually a bijection. Note that an isometry is automatically injective. It is surjective because it is the image of a complete thing under an isometry, hence complete, and its image contains $\phi_2(V)$ so its image must have the completion of $\phi_2(V)$ in $\overline{V}_2$ which is $\overline{V}_2$.

□

Proposition. A normed space V is complete if and only if every absolutely convergent series converges

Proof. Suppose V is complete and that $\sum_{n=1}^{\infty} \|x_n\| < \infty$. Then for $M > N$ we have

$$\left\| \sum_{n=N}^{M} x_n \right\| \leq \sum_{n=N}^{M} \|x_n\|$$

which goes to 0 as $n \to \infty$ so the sequence is Cauchy and hence converges.

Conversely, suppose that every absolutely convergent series is convergent and let (x_n) be a Cauchy sequence.

It is sufficient to find a subsequence x_{n_i} converging to some $x \in V$ because then

$$\|x_m - x\| \leq \|x_m - x_{n_i}\| + \|x_{n_i} - x\|$$

which goes to 0 as m, i go to ∞.

Now for each $i \in \mathbb{N}$ pick $n_i > n_{i-1}$ such that $m > n_i$ implies $\|x_m - x_{n_i}\| \leq 2^{-i}$, then $\sum_{i=1}^{\infty} \left\|x_{n_{i+1}} - x_{n_i}\right\| \leq 1$ so $x_{n_i} \to x$ for some $x \in V$ by the hypothesis that every absolutely convergent series is convergent.

□

1.5 Linear operators

Definition. Let X, Y be normed spaces and let T be a linear map $T : X \to Y$. Then T is said to be continuous if it is a continuous map in the metric spaces induced by the norms on X and Y.

Lemma. Let $T : X \to Y$ be a linear map, then the following are equivalent

1) T is continuous everywhere

2) T is continuous somewhere

3) T is bounded (meaning that there exists an R such that $\|Tx\|_Y \leq R \|x\|_X$)

Proof. $1 \implies 2$ is left as an exercise.

$2 \implies 3$: If T is continuous at x_0 then there exists a $\delta > 0$ such that $\|x - x_0\|_X < \delta$ implies $\|Tx - Tx_0\|_Y = \|T(x - x_0)\|_Y \leq 1$ Therefore by properties of norms, $\frac{1}{\delta}$ is an R that works.

$3 \implies 1$: (3) implies that $\|Tx\|_Y \leq R\|x\|_X$. Now let $\varepsilon > 0$ and take $\delta = \frac{\varepsilon}{R}$. Then if $\|x - x_0\| \leq \delta$ then we have $\|Tx - Tx_0\| = \|T(x - x_0)\| \leq \varepsilon$

□

Definition. Just like in $\mathbb{R}^n$, the infimum of all possible values of R in the proof above is called the operator norm.

Example. Let $X = l_1 = \{(x_1, x_2, ...) : \sum |x_i| < \infty\}$. Define $S : X \to X$ by

$$S(x_1, x_2, ...) = (x_2, x_3, ...)$$

Then it is easy to see that $\|S\| = 1$ since 1 is the largest possible value of a norm of the image of something with norm 1 under S.

Also, if we define

$$T(x_1, x_2, ...) = (0, x_1, x_2, ...)$$

then S is a left inverse to T but not a right inverse, so that property of matrices (that if there is one kind of inverse there is the other) fails.

Example. Let $X = C^1[0,1]$ i.e. the continuously differentiable functions on $[0,1]$ and use the supremum norm. Let $Y = C[0,1]$ with the supremum norm and define $T : X \to Y$ by $T(f) = f'$, then this is a linear map. The operator norm here is not finite, since an element of X given by $\sin(n\pi x)$ will have its norm scale by a factor of $n\pi$ after the transformation, and n can be made arbitrarily large.

Definition. Let X, Y be normed spaces. Then define

$$B(X,Y) = \{T : X \to Y : \|T\| < \infty\}$$

If $Y = \mathbb{R}$ or $Y = \mathbb{C}$ we call this the dual space of X, denoted X^*, and we call its elements Functionals.

We can easily check that this is a vector space and it is a normed space with the operator norm. If we have elements S and T then

$$\|S + T\| = \sup_{\|x\|=1} \|(S+T)x\| \leq \sup_{\|x\|=1} \|Sx\| + \sup_{\|x\|=1} \|Tx\| = \|S\| + \|T\|$$

The other rules are similarly easy to check.

Proposition. Let Y be a Banach space, then $B(X,Y)$ is also a Banach space.

Proof. Suppose that T_n is a Cauchy sequence in $B(X,Y)$ to that $\|T_m - T_n\| \to 0$ as $m, n \to \infty$. Then we need to find $T \in B(X,Y)$ such that $T_n \to T$ in the operator norm.

Let $x \in X$, then $(T_n x)_{n=1}^{\infty}$ is Cauchy in Y because

$$\|T_m x - T_n x\|_Y \leq \|T_m - T_n\| \|x\|$$

The RHS $\to 0$ as $m, n \to \infty$ and Y is a Banach space so there exists x^* such that $T_n x \to x^*$

Now define $Tx = x^*$, then T is linear as $T_n(cx) = cT_n(x)$ and

$$T_n(x+y) = T_n x + T_n y \to x^* + y^*$$

so $(x+y)^* = x^* + y^*$

It remains to check that $T_n \to T$ in the operator norm.

Let $\varepsilon > 0$ and suppose m is sufficiently large such that that $\|T_m - T_n\| \leq \varepsilon$ for all $n \geq m$. Then

$$\|T_m x - T_n x\|_Y \leq \varepsilon \|x\|_X$$

Letting $n \to \infty$ we get $\|T_m x - Tx\|_Y \leq \varepsilon \|x\|_X$

Therefore m is such that $\|T_m - T\| \leq \varepsilon$. Then done since ε was arbitrary. Also, $\|T\|$ is bounded since for n large enough we have

$$\|T\| \leq \|T_n\| + \|T - T_n\| \leq \|T_n\| + 1$$

□

Corollary. X^* is a Banach space for every X

Proposition. Let $1 < p < \infty$ and let q be conjugate to p such that $\frac{1}{p} + \frac{1}{q} = 1$, then l_p^* is isometrically isomorphic to l_q

Proof. Let $e_1 = (1,0,0,...)$ and $e_2 = (0,1,0,...)$ and $e_3 = (0,0,1,...)$ and so on. Note that this is not a basis for l_p since a basis means everything is a *finite* linear combination.

Now suppose that $\phi \in l_p^*$. Let $\phi(e_i) = y_i$. Then

$$\phi(x_1, ..., x_n, 0, 0, ...) = x_1 y_1 + x_2 y_2 + ... + x_n y_n$$

However, if $x = (x_1, x_2, ...) \in l_p$ then $(x_1, ..., x_n, 0, 0, ...) \to x$ in l_p (Note that this is false if $p = \infty$).

It follows that $\phi(x) = \phi(x_1, x_2, ...) = \sum_{i=1}^{\infty} x_i y_i$

It is equal to the sum since ϕ being continuous is equivalent to it being in l_p^*

It remains to show that $\phi \in l_p^* \iff (y_1, y_2, ...) \in l_q$ then we will have an isomorphism.

If $(y_1, y_2, ...) \in l_q$ then we define the map ϕ by this infinite sum. Holder's inequality implies

$$|\phi(x)| = \left|\sum x_i y_i\right| \le \|x\|_p \|y\|_q$$

This implies that $\phi \in l_p^*$

On the other hand, if $\phi \in l_p^*$ then there exists an M such that for all $(x_i) \in l_p$ we have

$$\left|\sum x_i y_i\right| \le M \|x\|_p$$

Fix N and let $x_i = \begin{cases} |y_i|^q y_i^{-1} & i \le N \\ 0 & i > N \end{cases}$

Then $(x_i) \in l_p$ so

$$\sum_{i=1}^{N} |y_i|^q \le M \left(\sum |y_i|^{(q-1)p}\right)^{\frac{1}{p}}$$

Therefore since $(q-1)p = q$ and $\frac{1}{p} = 1 - \frac{1}{q}$ we have

$$\left(\sum_{i=1}^{N} |y_i|^q\right)^{\frac{1}{q}} \le M$$

This gives an isomorphism between the dual space. We now need to show that it is isometric. We need to show that $\|y\|_{l_q} = \|\phi\|_{l_p^*}$. We can do this by splitting it into proving both $\le$ and $\ge$ after which both directions will follow with the same proof as the two directions we proved above.

□

Corollary. l_p is complete for $1 < p < \infty$ as it is a dual space and dual spaces are complete.

Note that this approach breaks down for l_1 (and in fact the statement is false as we may or may not see later in the course) because the sentence "$(x_1, ..., x_n, 0, 0, ...) \to x$ in l_p" is false in general if $p = \infty$

Definition. Let X, Y be normed spaces and let $T \in B(X, Y)$. Then the adjoint map is a map $T^* : Y^* \to X^*$ defined by $(T^* g)(x) = g(Tx)$ for all $g \in Y^*$, $x \in X$.

Then T^* is linear and it is bounded bounded because

$$\|T^* g(x)\| = \|g(Tx)\| \le \|g\| \|Tx\| \le \|g\| \|T\| \|x\|$$

Since this is true for every x we have $\|T^* g\| \le \|g\| \|T\|$ and therefore $\|T^*\| \le \|T\|$

We will see later that $\|T^*\| = \|T\|$.

2 The Hahn-Banach Theorem

2.1 Proof of the theorem

Definition. Let X be a real vector space and let $p : X \to \mathbb{R}$ be any function. Then p is said to be a convex functional if

1) $p(ax) = ap(x)$ for all $a \geq 0$

2) $p(tx + (1-t)y) \leq tp(x) + (1-t)p(y)$ for $t \in [0,1]$

Proposition. Any element of X^* is a convex functional

Proof. Follows from the fact that if $p \in X^*$ then $p(x+y) \leq p(x) + p(y)$ for any x, y, applied to tx and $(1-t)y$, and using linearity of p.

□

Lemma. If p is a convex functional then $p(a+b) \leq p(a) + p(b)$

Proof.

$$p(a+b) = 2p\left(\frac{a}{2} + \frac{b}{2}\right) \leq 2\left(\frac{1}{2}p(a) + \frac{1}{2}p(b)\right) = p(a) + p(b)$$

where the first 2 steps are by definition of a convex functional.

□

Lemma. Let X be a real vector space and let $p : X \to \mathbb{R}$ be a convex functional on X wish $M \subset X$ being a proper linear subspace and $f : M \to \mathbb{R}$ being a linear functional on M such that $f(x) \leq p(x)$ for all $x \in M$.

Let $x_0 \in X \setminus M$ and write $M' = M + \langle x_0 \rangle$meaning that any $z \in M'$ can be written in the form $z = y + tx_0$ with $t \in \mathbb{R}$ and $y \in M$.

Then there exists a linear functional $F : M' \to \mathbb{R}$ such that F restricted to M agrees with f and $F(x) \leq p(x)$ for all $x \in M'$

Proof. The only degree of freedom in this situation is to pick a value of c such that $F(x_0) = c$. When we do, $F(y + tx_0) = F(y) + tc$

For $t > 0$ we require $f(y) + tc \leq p(y + tx_0)$, which is equivalent to saying

$$c \leq p\left(\frac{y}{t} + x_0\right) - f\left(\frac{y}{t}\right)$$

For $t > 0$ let $t = -s$, then we require $f(y) - sc \leq p(y - sx_0)$, which is equivalent to saying

$$c \geq f\left(\frac{y}{s}\right) - p\left(\frac{y}{s} - x_0\right)$$

Such a c exists if

$$f(y_1) - p(y_1 - x_0) \leq p(y_2 + x_0) - f(y_2)$$

for all $y_1, y_2 \in M$

And indeed,

$$f(y_1) + f(y_2) = f(y_1 + y_2) \leq p(y_1 + y_2) \leq p(y_1 - x_0) + p(y_2 + x_0)$$

□

Recall from the Logic and Set theory course that Zorn's lemma states that every non-empty partially ordered set with the property that every chain has a least upper bound has a maximal element. It is perhaps surprising that it is coming up here, but we are going to use it shortly.

Lemma. Let X be a real vector space and let $p : X \to \mathbb{R}$ be a convex functional on X wish $M \subset X$ being a proper linear subspace and $f : M \to \mathbb{R}$ being a linear functional on M such that $f(x) \leq p(x)$ for all $x \in M$.

Then there exists $F : X \to \mathbb{R}$ linear such that F restricted to M agrees with f and $F(x) \leq p(x)$ for all $x \in X$

Proof. Define

$$\mathcal{F} = \left\{ \left(\widetilde{M}, \widetilde{f} \right) \right\}$$

where $\widetilde{M}$ is a linear subspace such that $\widetilde{M} \supset M$ and $\widetilde{f} : \widetilde{M} \to \mathbb{R}$ is linear and $\widetilde{f} = f$ on M and $\widetilde{f} \leq p$ everywhere.

Define a partial ordering by $\left(\widetilde{M}, \widetilde{f} \right) \leq \left(\widetilde{N}, \widetilde{g} \right)$ if $\widetilde{M} \subset \widetilde{N}$ and $\widetilde{g}|_{\widetilde{M}} = \widetilde{f}$

$\mathcal{F}$ is non-empty since $(M, f) \in \mathcal{F}$. If $\{(M_i, g_i) : i \in I\}$is a totally ordered subset then it has an upper bound given by $\widetilde{M} = \bigcup_{i \in I} M_i$ and $\tilde{g}(x) = g_i(x)$ if $x \in M_i$.

By Zorn's lemma, $\mathcal{F}$ has a maximal element (N, F). We must have $N = X$ as otherwise we could pick some $x_0 \in X \setminus N$ and extend F by the previous lemma, contradicting minimality.

$\square$

Definition. Let X be a vector space over $\mathbb{R}$ or $\mathbb{C}$. A map $q : X \to \mathbb{R}$ is called a seminorm if

1. $q(ax) = |a| q(x)$ for all $x \in X$ and a in $\mathbb{R}$ or $\mathbb{C}$

2. $q(x + y) \leq q(x) + q(y)$

We drop the condition about when the norm is 0.

Theorem. (Hahn-Banach Theorem) Let X be a vector space over $\mathbb{K}$ which is either $\mathbb{R}$ or $\mathbb{C}$ and let $q : X \to \mathbb{R}$ be a seminorm on X and let $M \subset X$ be a linear subspace and let $f : M \to \mathbb{K}$ be linear such that $|f(x)| \leq q(x)$ for all $x \in M$

Then there exists $F : X \to \mathbb{K}$ linear with $F|_M = f$ and $|F(x)| \leq q(x)$ for all $x \in X$

Proof. For $\mathbb{K} = \mathbb{R}$, apply the previous lemma. We know there is an F satisfying the property we want with $F(x) \leq q(x)$ and we want to show that $-F(x) \leq q(x)$. But we have

$$-F(x) = F(-x) \leq q(-x) = q(x)$$

For $\mathbb{K} = \mathbb{C}$ consider M, X as real vector spaces. Then $\mathrm{Re}(f) : M \to \mathbb{R}$ is linear with respect to $\mathbb{R}$ and $|\mathrm{Re}(f)| \leq |f| \leq |q|$ on M so there exists $g : X \to \mathbb{R}$ which is linear with respect to $\mathbb{R}$ and agrees with $\mathrm{Re}(f)$ on M and has $|g(x)| \leq q(x)$ for all $x \in X$.

Let $F(x) = g(x) - ig(ix)$. Then

$$F(ix) = g(ix) - ig(-x) = g(ix) + ig(x) = iF(x)$$

Therefore F is linear with respect to $\mathbb{C}$.

Also, if $x \in M$ then we have

$$\mathrm{Re}(F(x)) = g(x) = \mathrm{Re}(f(x))$$

and

$$\mathrm{Im}(F(x)) = -g(ix) = -\mathrm{Re}(f(ix)) = -\mathrm{Re}(if(x)) = \mathrm{Im}(f(x))$$

Thus F, f agree on M. We also have for all $x \in X$ that if $\theta = -\arg(F(x))$ then

$$|F(x)| = \left|e^{i\theta}F(x)\right| = \left|F\left(e^{i\theta}x\right)\right| = \left|g\left(e^{i\theta}x\right)\right| \leq q\left(e^{i\theta}x\right) = q(x)$$

The last equality is because q is a seminorm.

□

2.2 Consequences of the theorem

Here, $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$ and X is a normed space over $\mathbb{K}$

Proposition. Let M be a linear subspace of X and let $f \in M^*$, then there exists $F \in X^*$ such that $F = f$ on M and $\|F\| = \|f\|$.

Proof. If $q(x) = \|f\| \|x\|$ then q is a seminorm and by Hahn-Banach there exists $F : X \to \mathbb{K}$ linear that agrees with f on M such that $|F(x)| \leq \|f\| \|x\|$ for all $x \in X$ so $\|F\| \leq \|f\|$, but since $F|_M = f$ we have $\|F\| = \|f\|$.

□

Proposition. Let $x_0 \in X \setminus \{0\}$, then there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x_0) = \|x_0\|$

Proof. Let $M = \langle x_0 \rangle$. Then the function $\tilde{f}(\lambda x_0) = \lambda \|x_0\|$ for $\lambda \in \mathbb{K}$ has $\left\|\tilde{f}\right\| = 1$. By Hahn-Banach (in fact by the previous proposition) there is $f \in X^*$ with $\|f\| = 1$ and $f|_M = \tilde{f}$ so $f(x_0) = \|x_0\|$

□

Remark. If $x_0 = 0$ the proposition is true. Just pick any $y \neq 0$ and apply the proposition to get an f with operator norm 1, then this f works.

Proposition. Let Z be a closed linear subspace of X and let $y \in X \setminus Z$ and let $d = \operatorname{dist}(y, z) = \inf_{y \in Z} \|y - z\|$. Then there exists $F \in X^*$ such that $\|F\| = 1$, $F|_Z = 0$ and $F(y) = d$

Proof. Note that $d > 0$ because Z is closed. Let $M = Z + \langle y \rangle$ and let $f : M \to \mathbb{K}$ be defined by $f(z + ty) = td$. Then f is linear and

$$\|f\| = \sup_{z \in Z, t \in \mathbb{K} \setminus 0} \frac{|f(z + ty)|}{\|z + ty\|} = \sup_{z \in Z, t \in \mathbb{K} \setminus 0} \frac{|t| d}{|t| \left\|\frac{z}{t} + y\right\|} = \sup_{z \in Z, t \in \mathbb{K} \setminus 0} \frac{d}{\left\|\frac{z}{t} + y\right\|} = \frac{d}{d} = 1$$

Thus $\|f\| = 1$. By the previous proposition there exists $F : X \to \mathbb{R}$ with $\|F\| = 1$ and $F|_M = f$. Therefore this F satisfies all conditions in the proposition.

□

Proposition. For all $x, y \in X$ with $x \neq y$ there exists $f \in X^*$ such that $f(x) \neq f(y)$. We say X^* separates the points of X.

Proof. If $x \neq y$ then $w = x - y \neq 0$ so by the previous proposition there exists $F \in X^*$ such that $\|F\| = 1$ and $F(x) - F(y) = F(x - y) = \|x - y\| > 0$ (since the zero point is a closed subspace).

□

Proposition. For all $x \in X$, $\|x\| = \sup_{f \in X^*, \|f\| \leq 1} |f(x)|$

Proof. We know that $|f(x)| \leq \|f\| \|x\| \leq \|x\|$ for all x and all $f \in X^*, \|f\| \leq 1$. On the other hand, if $x \in X \setminus 0$, by the previous proposition there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x) = \|x\|$ and therefore $\sup_{f \in X^*, \|f\| \leq 1} |f(x)| \geq \|x\|$.

□

Proposition. Let $T \in B(X, Y)$ and let $T^* : Y^* \to X^*$ be the corresponsing map (so that $T^* \circ f(x) = f \circ T(x)$). Then $T^* \in B(Y^*, X^*)$ and $\|T^*\| = \|T\|$

Proof. We showed earlier that $T^* \in B(Y^*, X^*)$ and $\|T^*\| \leq \|T\|$. We aim here to show that equality occurs.

Given $\varepsilon > 0$ it is possible to find $x_0 \in X$ with $\|x_0\| \leq 1$ and $\|T\| \leq \|T(x_0)\| + \varepsilon$. By an earlier proposition there exists $f \in Y^*$ such that $\|f\| = 1$ and $f(T(x_0)) = \|T(x_0)\|$. Then

$$\|T\| - \varepsilon \leq f(Tx_0) = (T^* f)\, x_0$$

Thus, $\|T^* f\| \geq \|T\| - \varepsilon$ and thus $\|T^*\| \geq \|T\| - \varepsilon$. This is true for all ε so $\|T^*\| \geq \|T\|$ as required

□

Definition. A normed space X is said to be separable if it has a countable dense subset

Remark. We know that the above definition is a useful property for a space to have considering how much we have used it in the past when saying "ennumerate the rational points" in a proof.

Example. l_∞ is NOT separable, because there is an uncountable subset where each element differs from each other element by at least 1. Specifically, this is the set of all sequences of elements in $\{0, 1\}$. Therefore there are uncountably many disjoint open intervals in l_∞ and a countable dense set would have to go into all of them which is impossible.

Theorem. Let X be a normed space, then if X^* is separable then X is separable.

Proof. Let $(f_k)_{k \in \mathbb{N}}$ be a dense subset of X^*. Then for all $k \in \mathbb{N}$ there exists $x_k \in X$ with $\|x_k\| = 1$ such that $|f_k(x_k)| \geq \frac{1}{2} \|f_k\|$. Let A be the set of all finite linear combinations of the $\{x_k\}$ with rational coefficients, then by general properties of countable sets A is countable.

We want to show that A is dense in X. We note that $\overline{A}$ is the space of combinations of x_k with real coefficients, as well as the limit points of this space, so this is actually a vector space. Suppose that $\overline{A} \neq X$. Then, by a proposition above there exists $f \in X^*$ such that $f \neq 0$ and $f|_{\overline{A}} = 0$.

Thus, by density there is a subsequence k_j such that $f_{k_j} \to f$ in X^* (since for each j we can pick $k_j > k_{j-1}$ such that $\left\|f_{k_j} - f\right\| < \frac{1}{j}$ as density forces the existence of such a k_j for each j and thus infinitely many such k_j for each j).

Then $\left\|f_{k_j} - f\right\| \to 0$ but on the other hand

$$\left\|f_{k_j} - f\right\| \geq \left|f_{k_j}(x_{k_j}) - f(x_{k_j})\right| = \left|f_{k_j}(x_{k_j})\right| \geq \frac{1}{2}\left\|f_{k_j}\right\|$$

This means $\left\|f_{k_j}\right\| \to 0$ as $j \to \infty$ which implies $f = 0$. This is a contradiction so done.

□

Definition. Let X be a normed space. Then we define $(X^*)^*$ to be its second dual. Note that this is always a Banach space. Define $\phi_X : X \to X^{**}$ as follows:

Note that ϕ_X sends $x \in X$ to a linear function from $X^* \to \mathbb{K}$ where X^* is itself the set of linear functions from $X \to \mathbb{K}$. We define $\phi_X(x)$ to be a function from $X^* \to \mathbb{K}$ such that $\phi_X(x)(f) = f(x)$.

Proposition. ϕ_X is linear and an isometry (Note that since we are about to show that it is an isometry, we also know that it is injective)

Proof. Linearity follows from linearity of each f.

To show that $\|\phi_X(x)\| = \|x\|$ we will show that $\|\phi_X(x)\| \leq \|x\|$ and $\|\phi_X(x)\| \geq \|x\|$.

We know

$$|\phi_X(x)(f)| = |f(x)| \leq \|f\| \|x\|$$

Therefore

$$\sup_{f \neq 0} \frac{|\phi_X(x)(f)|}{\|f\|} \leq \|x\|$$

Therefore $\|\phi_X(x)\| \leq \|x\|$.

On the other hand, if $x \neq 0$ then by an earlier proposition there exists $\tilde{f} \in X^*$ such that $\left\|\tilde{f}\right\| = 1$ and $\left\|\tilde{f}(x)\right\| = \|x\|$. Therefore, $|\phi_X(x)(f)| = \|x\|$ and thus

$$\frac{\left|\phi_X(x)\left(\tilde{f}\right)\right|}{\left\|\tilde{f}\right\|} = \|x\|$$

Thus $\|\phi_X(x)\| \geq \|x\|$.

$\square$

Definition. A normed space X is called reflexive if ϕ_X is surjective and thus defines an isometric isomorphism between X and X^{**}.

Remark. Every reflexive space is complete. However, a Banach space need not be reflexive.

Theorem. Let X be a reflexive Banach space and let $M \subset X$ be a closed linear subspace, then M is reflexive.

Proof. We have $\phi_X : X \to X^{**}$ and $\phi_M : M \to M^{**}$. We can define a map $j : M \to X$ such that $j(m) = m$ and then we have the corresponding map $j^* : M^* \to X^*$ such that $j^*(f)(m) = f(j(m)) = f(m)$. Essentially what I am saying is that for $f \in M^*$ we have $j^*(f) = f|_M$.

We also have $j^{**} : M^{**} \to X^{**}$ such that

$$j^{**}(m^{**})(f) = m^{**}(j^*(f)) = m^{**}(f|_M)$$

Now let $m^{**} \in M^{**}$. To show that ϕ_M is surjective we want $x \in M$ such that $\phi_M(x) = m^{**}$.

Since X is reflexive there exists $x \in X$ such that $\phi_X(x) = j^{**}(m^{**})$. If $x \notin M$ then by an earlier proposition there exists $x^* \in X^*$ such that $x^*|_M = 0$ and $x^*(x) = 1$. It is here we use the fact that M is closed. We have

$$1 = x^*(x) = \phi_X(x)(x^*) = j^{**}(m^{**})(x^*) = m^{**}(j^*(x^*)) = m^{**}(x^*|_M) = 0$$

This is a contradiction so $x \in M$. We will be done if we can show that $\phi_M(x) = m^{**}$.

Now $\phi_X(j(m))$ is in X^{**} so it is a function acting on X^*. We have

$$\phi_X(j(m))(x^*) = \phi_X(m)(x^*) = x^*(m)$$

But $j^{**}(\phi_M(m))$ is also in X^{**} and we have, by definition of the double dual map,

$$j^{**}(\phi_M(m))(x^*) = \phi_M(m)(j^*(x^*)) = \phi_M(m)(x^*|_M) = x^*(m)$$

This means it follows that the diagram in Figure 1 makes sense:

$$\begin{array}{ccc} X & \xrightarrow{\varphi_X} & X^{**} \\ j \uparrow & & \uparrow j^{**} \\ M & \xrightarrow{\varphi_M} & M^{**} \end{array}$$

Figure 1

For arbitrary $m^* \in M^*$, let $x^* \in X^*$ be such that $x^*|_M = m^*$ (possible by an earlier proposition). Then

$$\phi_M(x)(m^*) = m^*(x) = x^*(x) = \phi_X(x)(x^*) = j^{**}(m^{**})(x^*)$$
$$= m^{**}(j^*(x^*)) = m^{**}(x^*|_M) = m^{**}(m^*)$$

Thus $\phi_M(x) = m^{**}$.

□

Theorem. Let X be a Banach space, then X is reflexive if and only if X^* is reflexive.

Proof. **Direction 1.** Suppose that X is reflexive, then we want to show that $\phi_{X^*} : X^* \to X^{***}$ is surjective.

Let $x^{***} \in X^{***}$ and define $x^* = x^{***} \circ \phi_X$ as a function that takes X to X^{**} then to $\mathbb{R}$. Then x^* is linear and continuous (composition of continuous functions) and thus it is in X^*.

For any $x^{**} \in X^{**}$ we have $x^{**} = \phi_X(x)$ for some $x \in X$ because X is reflexive. Then

$$\phi_{X^*}(x^*)(x^{**}) = x^{**}(x^*) = \phi_X(x)x^{**} = x^*(x) = x^{***}(\phi_X(x)) = x^{***}(x^{**})$$

This means that $\phi_{X^*}(x^*) = x^{***}$. Since x^{***} was arbitrary it follows that $\phi_{X^*}$is surjective.

Direction 2. Now suppose X^* is reflexive. Then this means that $\phi_{X^*} : X^* \to X^{***}$ is surjective and our goal is to show that $\phi_X : X \to X^{**}$ is also surjective.

Assume for contradiction that ϕ_X is not surjective, then $\phi_X(X)$ is a proper closed subspace of X^{**}. By an earlier proposition, there exists a non-zero continuous linear functional on X^{**} that vanishes exactly on $\phi_X(X)$, call this x^{***}.

By reflexivity of X^* there exists an $x^* \in X^*$ such that $\phi_{X^*}(x^*) = x^{***}$. For any $x \in X$ we have

$$x^{***}(\phi_X(x)) = \phi_{X^*}(x^*)(\phi_X(x)) = \phi_X(x)(x^*) = x^*(x)$$

Since x^{***} vanishes when applied to anything in $\phi_X(X)$ it follows that $x^*(x) = 0$ for all $x \in X$. This means that $x^* = 0$, so $x^{***} = \phi_{X^*}(0) = 0$ which is a contradiction.

□

Note that l_1 is separable with the countable dense subsets being all sequences of rational numbers that are eventually 0.

Example. l_p is reflexive for $1 < p < \infty$ because $l_p^{**} \cong l_q^* \cong l_p$ if q is conjugate to p. However, l_1 and l_∞ are not reflexive, since if l_1 were reflexive then $l_1 \cong l_1^{**}$ implies l_1^{**} is separable which implies l_1^* is separable which is a contradiction. This also proves that the dual of l_∞ is not l_1.

3 The Baire Category Theorem

3.1 Proof of the theorem

The next big theorem of the course is the Baire Category Theorem and we will see that this has many interesting applications.

Recall that in a Topological space, a set is dense if it intersects every open set or equivalently if its closure is the whole space. In a Metric space, a set is dense if it is dense with respect to the induced topology.

Definition. The diameter of a set in a metric space is the supremum of all distances between any 2 points in the set.

Lemma. Let X be a complete metric space and let $F_1 \supseteq F_2 \supseteq F_3...$ be a nested sequence of closed, non-empty subsets of X with $\operatorname{diam}(F_n) \to 0$. Then the F_n's have non-empty intersection.

Proof. Pick $x_n \in F_n$ for each i. Since $d(x_n, x_m) \leq \text{diam}\left(F_{\min(m,n)}\right) \to 0$ as $n, m \to \infty$, x_n is a Cauchy sequence. By completeness, $x_n \to x^*$ for some x^*. For each m, the sequence $x_m, x_{m+1}, x_{m+2}, ...$ lies in F_m and thus so does its limit by closure. Thus $x^* \in F_m$ for each F_m so we conclude that the intersection is indeed non-empty

$\square$

Theorem. (Baire Category Theorem v1) Let $(G_n)_{n=1}^{\infty}$ be a sequence of open dense subsets in a complete metric space, then $\bigcap_{n=1}^{\infty} G_n$ is dense

Proof. Consider some ball $B_{r_0}(x_0)$, then we want to prove that a point in $\bigcap_{n=1}^{\infty} G_n$ lives in this ball.

Since G_1 is open and dense, $B_{r_0}(x_0)$ meets G_1 in an open set, and therefore, $B_{r_0}(x_0) \cap G_1$ is open so contains an open ball. In fact, it contains a closed ball of the form $\overline{B_{r_1}(x_1)}$ with $r_1 > 0$.

Now $\overline{B_{r_1}(x_1)}$ meets G_2 so we can find a closed ball $\overline{B_{r_2}(x_2)}$ in $\overline{B_{r_1}(x_1)} \bigcap G_2$. It is possible to keep doing this process in such a way that $r_n \to 0$.

Thus, the closed balls have non-empty intersection by the lemma so we conclude that there is a point in all G_n in the arbitrary open ball we chose at the start. Thus, $\bigcap_{n=1}^{\infty} G_n$ is dense.

$\square$

Theorem. (Baire Category Theorem v2) Let X be a complete metric space and let $(F_n)_{n=1}^{\infty}$ be a sequence of closed sets with union equal to X. Then at least one F_n has non-empty interior. That is, some open ball is contained in it.

Proof. Take $G_n = X \setminus F_n$. By the contrapositive of Baire Category Theorem, since the G_n's have empty intersection, we must have that at least one of G_n is not dense, that is, said G_n does not intersect some open interval, which thus must be in said F_n.

$\square$

Definition. A set is said to be meagre, or of first category, if it is a countable union of closed sets with empty interior, and of second category otherwise. Then Baire Category Theorem says a complete metric space is of second category.

3.2 Consequences of the theorem

Lemma. if $x_i \to x$ and f_i are continuous and converge uniformly to f then $f_i(x_i) \to f(x)$

Proof. We know f is continuous by standard analysis. Then we have

$$|f_i(x_i) - f(x)| \leq |f_i(x_i) - f(x_i)| + |f(x_i) - f(x)|$$

By continuity of f the second term tends to 0 and by uniform convergence the first term tends to 0.

$\square$

Proposition. There exists a continuous function on $[0, 1]$ that is differentiable nowhere in $(0, 1)$

In the misc results section of this website, we construct such a function and prove it has this property. However, in the following proof we will be able to prove such a function exists without constructing it explicitly.

Proof. Define S_n to be the set of all $f \in C[0, 1]$ such that there exists some $x \in (0, 1)$ such that $\frac{|f(x)-f(y)|}{|x-y|} \leq n$ whenever $0 < |x - y| \leq \frac{1}{n}$.

If f is differentiable at x then f lies in some S_n. If the theorem is false then $\bigcup_{n=1}^{\infty} S_n = C[0,1]$. Therefore it suffices to show that each S_n is closed and has empty interior by the Baire Category Theorem.

Note that the metric we will be working with here is the infinity norm.

Step 1. Closed

Suppose $(f_i)_{i=1}^{\infty}$ is a sequence in S_n such that $f_i \to f$ for some $f \in C[0,1]$ (We are using the infinity norm so this is equivalent to Uniform convergence). We aim to show that $f \in S_n$.

For each i, there is x_i such that $\frac{|f_i(x_i)-f_i(y)|}{|x_i-y|} \leq n$ whenever $0 < |x_i - y| \leq \frac{1}{n}$. Also, x_i is a bounded sequence so it has a convergent subsequence x_{i_j} in $[0,1]$, we will call the limit x.

Suppose that $0 < |x-y| < \frac{1}{n}$, then we have $0 < \left|x_{i_j} - y\right| < \frac{1}{n}$ for j sufficiently large, so by the lemma above we have

$$\frac{|f(x) - f(y)|}{|x-y|} = \lim_{i\to\infty} \frac{\left|f_{i_j}\left(x_{i_j}\right) - f_{i_j}(y)\right|}{\left|x_{i_j} - y\right|} \leq n$$

Step 2. Empty interior

We want to prove that for any continuous $f_0 \in C[0,1]$, for all $\varepsilon > 0$ there is some $f \in C[0,1]$ not in S_n with $\|f - f_0\|_{\infty} < \varepsilon$.

The way to do this is basically Figure 2, where f_0 is the function in black and f will look like the function in red.

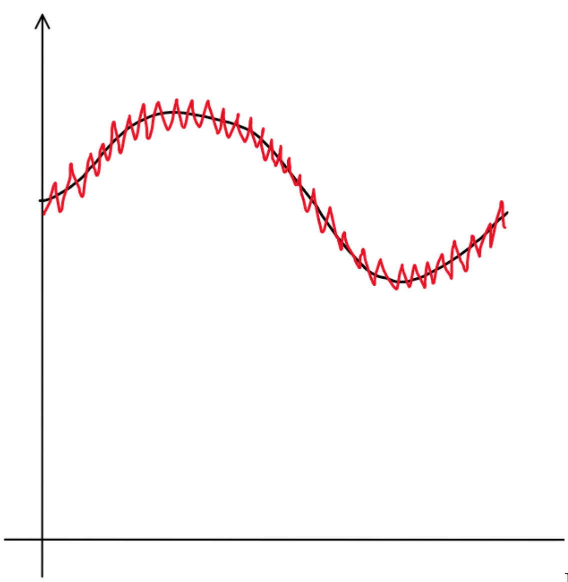

Figure 2

We will define a piecewise linear function f_1 with $\|f_1 - f_0\| < \frac{\varepsilon}{2}$. We will define f_1 by making M large and making f_1 agree with f at i/M for each integer i and interpolate between these points. By uniform continuity, we can find M such that when $|x-y| < \frac{1}{M}$ we have $f_0(x) - f_0(y) < \frac{\varepsilon}{4}$. It is easy to see that this implies $\|f_1 - f_0\| < \frac{\varepsilon}{2}$ for this choice of M.

Now let $M' > M$. Define $g\left(\frac{i}{M'}\right) = (-1)^i$ and interpolate linearly. The slope of g at every point where it is differentiable is $2M'$. What we then do is we set $f = f_1 + \frac{\varepsilon}{2} g$ with M' set large enough so that it is greater by n than the largest slope attained by f_1, so that f is not in S_n as required.

$\square$

Theorem. (Uniform boundedness principle) Let X be a complete metric space and let $\mathcal{F}$ be a collection of continuous real-valued functions on X. Suppose that for each $x \in X$, $\sup_{f\in\mathcal{F}} |f(x)| < \infty$.

Then there is a ball $B_\varepsilon(x_0)$, $\varepsilon > 0$, such that

$$\sup_{f\in\mathcal{F}} \sup_{x\in B_\varepsilon(x_0)} |f(x)| < \infty$$

Proof. Define

$$S_n = \bigcap_{f\in\mathcal{F}} \{x \in X : |f(x)| \le n\}$$

Then this is an intersection of closed sets so it is closed.

Also, $\bigcup_{n=1}^{\infty} S_n = X$, since $x \in S_n$ whenever $n > \sup_{f\in\mathcal{F}} |f(x)|$.

Since X is complete, at least one S_n has non-empty interior by the Baire Category Theorem.

□

Theorem. (Banach-Steinhaus uniform boundedness principle) Suppose X is a Banach space and $\mathcal{T}$ is a family of bounded linear functions from X to some other normed space Y. Suppose $\sup_{T\in\mathcal{T}} \|Tx\| < \infty$ for all $x \in X$. Then $\sup_{T\in\mathcal{T}} \|T\| < \infty$

Proof. Apply the previous theorem to the functions $x \mapsto \|Tx\|$ for $T \in \mathcal{T}$. Then this is a family of continuous functions on a complete metric space. Therefore there is a ball $B_\varepsilon(x_0)$ with $\varepsilon > 0$ such that

$$\sup_{T\in\mathcal{T}} \sup_{x\in B_\varepsilon(x_0)} \|Tx\| < \infty$$

Call this supremum M.

Then if $z \in X$ has $\|z\| < \frac{\varepsilon}{2}$ then $\|T(x_0 + z)\| \le M$ and $\|T(x_0 - z)\| \le M$. Therefore

$$\|Tz\| = \frac{1}{2}\|T(x_0 + z) - T(x_0 - z)\| \le M$$

Therefore we have $\|Ty\| \le \frac{2M}{\varepsilon}$ for all T and all y with $\|y\| \le 1$

Therefore $\sup_{T\in\mathcal{T}} \|T\| < \frac{2M}{\varepsilon}$

□

Theorem. (Open mapping theorem, Linear analysis version not to be confused with Complex analysis version) Let $T : X \to Y$ be a surjective bounded linear operator between 2 Banach spaces X, Y. Then T maps open sets to open sets.

Proof. It suffices to show that $T(B_1(0))$ is open, since

$$T(B_\varepsilon(x)) = T(x) + \varepsilon T(B_1(0))$$

would therefore also be open.

Step 1. We will prove that $\overline{T(B_1(0))}$ contains some open ball centered about the origin in Y, and in step 2 we will remove the closure.

To do this, T being surjective implies that $Y = \bigcup_{n=1}^{\infty} T(B_n(0))$. But $T(B_n(0)) = nT(B_1(0))$ so we have $Y = \bigcup_{n=1}^{\infty} n\overline{T(B_1(0))}$.

By the Baire Category Theorem it follows that one of the $\overline{T(B_1(0))}$ has non-empty interior, and thus contains an open ball.

Recall that open balls are convex, then $\overline{T(B_1(0))}$ is therefore symmetric about the origin and convex (since it is the closure of the image of a convex set under a linear map, and this is convex because a convex combination of 2 limit points is the limit point of the convex combinations and thus in the set).

Since $\overline{T(B_1(0))}$ is open it has some $B_\delta(y_0)$ but then it also has $B_\delta(-y_0)$. Therefore if $\|z\| < \delta$ then $y_0 + z$ and $-y_0 + z$ are in this and by convexity so is z.

Step 2. We will now try to remove the closure.

For notational symmetry, assume $\delta = 1$ as we can replace T by $\frac{1}{\delta}T$.

Now we will write $B_X(1)$ as the open ball in X and $B_Y(1)$ as the open ball in Y to avoid ambiguity. We will show that $B_Y(1) \subseteq \overline{T(B_X(1))}$ implies $B_Y(1) \subseteq T(B_X(1))$.

Fix $\varepsilon > 0$. Note that by $B_Y(1) \subseteq \overline{T(B_X(1))}$, take y and let $r = \|y\|_Y$. By scaling, $B_Y(r) \subseteq \overline{T(B_X(r))}$ so $\overline{B_Y(r)} \subseteq \overline{T(B_X(r))}$ so $y \subseteq \overline{T(B_X(r))}$. Therefore given any $\varepsilon > 0$ there exists $\hat{y} \in T(B_X(r))$ with $\|y - \hat{y}\|_Y < \varepsilon$, so there exists $x \in B_X(r)$ with $\|y - Tx\|_Y < \varepsilon$. Therefore for any y there is an x such that $\|x\|_X \leq \|y\|_Y$ and $y = Tx + y'$ with $\|y'\|_Y \leq \varepsilon$.

Now let $y \in B_Y(1)$ and set $y_1 = y$.

Find x_1 with $\|x_1\| \leq \|y_1\|$ such that $y_1 = Tx_1 + y_2$ with $\|y_2\| \leq \frac{1-\|y\|}{2^2}$.

Find x_2 with $\|x_2\| \leq \|y_2\|$ such that $y_2 = Tx_2 + y_3$ with $\|y_3\| \leq \frac{1-\|y\|}{2^3}$.

Do this repeatedly and define $x = x_1 + x_2 + x_3 + ...$ and this sum makes sense since we are in a Banach space and the sequence of partial sums is Cauchy by the triangle inequality and the fact that $\sum \frac{1-\|y\|}{2^n}$ converges.

Since T is bounded and therefore continuous, by construction we have

$$Tx = \lim_{n\to\infty} T(x_1 + ... + x_n) = \lim_{n\to\infty} (y - y_{n+1}) = y$$

Also by the triangle inequality we have

$$\|x\| \leq \|x_1\| + \|x_2\| + ... \leq \|y_1\| + \|y_2\| + ... < \|y_1\| + 1 - \|y\| = 1$$

Now we have found $x \in B_X(1)$ such that $Tx = y$. Since y was arbitrary this implies that $B_Y(1) \subseteq T(B_X(1))$

□

Corollary. (Identity/Inversion principle) Let X and Y be Banach spaces and let $T : X \to Y$ be bounded, linear and bijective. Then T^{-1} is a bounded linear operator.

Proof. Clearly it is linear. Boundedness follows from the fact that if $U \subseteq X$ is open then $\left(T^{-1}\right)^{-1} U = TU$ is open by the previous theorem, so pre-images of open sets are open under T^{-1} which is therefore continuous and thus bounded.

□

Lemma. If X, Y are normed spaces and we assign the norm to $X \times Y$ given by $\|(x,y)\|_{X\times Y} = \|x\|_X + \|y\|_Y$ then the induced topology is the product topology.

Proof. We just need to prove that an open ball in the induced topology is a union of open rectangles in the product topology, and vice versa.

To do this, note that if we have an open rectangle R in the product topology then about any point (x, y) in R there is some open rectangle of size $\varepsilon \times \varepsilon$ around it contained in R. Then every point with $\|a_1 - x\|_X + \|a_2 - y\|_Y < \varepsilon$ is contained in this open rectangle.

Conversely if we have an open ball B in the induced topology then about any point (x, y) in B there is some open ball around it containing all points (a_1, a_2) with $\|a_1 - x\|_X + \|a_2 - y\|_Y < \varepsilon$ since open balls are open. Then this contains the open rectangle of size $\frac{\varepsilon}{2} \times \frac{\varepsilon}{2}$ in the product topology.

□

Theorem. (Closed graph theorem) Let X, Y be Banach spaces and let $T : X \to Y$ be a linear operator. Then T is bounded if and only if the graph $\Gamma = \{(x, Tx) : x \in X\}$ is closed in the product topology on $X \times Y$.

Proof. Note that clearly if the linear map is bounded it is continuous and therefore the graph is closed.

Conversely, suppose that the graph is closed in the product topology, then it is closed in the induced topology by the norm above. $X \times Y$ is a Banach space and Γ is a closed linear subspace of it and thus also a Banach space with the same norm.

Now consider the map $\pi : \Gamma \to X$ defined by projection, namely $\pi(x, y) = x$. This is linear, bounded and bijective. Therefore by the identity/inversion principle the inverse $\pi^{-1} : X \to \Gamma$ is a bounded linear operator. Therefore there is a constant C such that

$$\|x\|_X + \|Tx\|_Y = \|(x, Tx)\|_{X\times Y} = \left\|\pi^{-1}(x)\right\|_{X\times Y} \leq C\,\|X\|_x$$

Therefore $\|Tx\|_Y \leq (C-1)\,\|x\|_X$. This proves T is bounded if the graph is closed.

□

Theorem. The pointwise limit of a sequence of continuous functions on any interval $I \subseteq \mathbb{R}$ has a point of continuity

Proof. For any function h and any set A define the oscillation of h as

$$w(h, A) = \sup_{x,y\in A} |h(x) - h(y)|$$

Let $g_1, g_2, ...$ be a sequence of continuous functions converging pointwise to some function h.

Let O_n be the union of all open sets $W \subseteq I$ such that $w(h, W) < \frac{1}{n}$. Then O_n is open and if $x \in \bigcap_{n=1}^{\infty} O_n$ then for every n, x is contained in some open neighbourhood where the oscillation of h is $< \frac{1}{n}$. This means that $\bigcap_{n=1}^{\infty} O_n$ is exactly the set of continuity points of h.

It suffices to show each O_n is dense by the Baire Category Theorem.

Let $U \subseteq I$ be an arbitrary non-empty open interval. We need to find an open set $W \subseteq U$ such that $w(h, W) < \frac{1}{n}$. Finding a W with this property would complete the proof because U has been chosen arbitrarily.

Fix $\varepsilon = \frac{1}{3n}$. For each $m \in \mathbb{N}$ define the set

$$F_m = \{x \in \mathbb{R} : \forall i, j \geq m\, |g_i(x) - g_j(x)| \leq \varepsilon\}$$

Because each g_k is continuous the condition $|g_i(x) - g_j(x)| \leq \varepsilon$ defines a closed set. This means F_m is an intersection of closed sets so it is also closed.

Since $g_k \to h$, the sequence $g_k(x)$ is Cauchy for every x. Thus, every $x \in I$ belongs to F_m for some m so

$$I = \bigcup_{m=1}^{\infty} F_m$$

Now let K be a closed interval with $K \subseteq U$. Then we have

$$K = \bigcup_{m=1}^{\infty} (F_m \cap K)$$

K is a complete metric space so by the Baire Category Theorem, at least one of the sets $F_m \cap K$ must have a non-empty interior. Therefore, there exists an integer m and a non-empty open set $V \subseteq K \subseteq U$ such that $V \subseteq F_m$.

For any $x \in V$ and $i \geq m$, taking the limit as $j \to \infty$ in the definition of F_m gives

$$|g_i(x) - h(x)| \leq \varepsilon$$

Because g_m is continuous there exists a non-empty open set $W \subseteq V$ such that the oscillation of g_m on W is $< \varepsilon$. We will show that this is the W we want. We can bound the oscillation of h on W by

$$|h(x) - h(y)| \leq |h(x) - g_m(x)| + |g_m(x) + g_m(y)| + |g_m(y) - h(y)|$$

Since $W \subseteq F_m$ the first and third terms are at most ε and we know the second term is $< \varepsilon$ by definition of W. Therefore

$$|h(x) - h(y)| < 3\varepsilon = \frac{1}{n} \text{ on } W$$

This means W has exactly the property we needed.

□

Corollary. Every derivative has a point of continuity in every interval

Proof. A derivative of a differentiable function f is the limit of the continuous functions $n\left(f\left(x + \frac{1}{n}\right) - f(x)\right)$ then the result follows by the previous theorem.

□

Definition. A subset of $\mathbb{R}$ is called perfect if it is closed and has no isolated points.

Example. Closed intervals are perfect, the cantor set is perfect, but $\left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup 0$ is not because it has isolated points.

Corollary. Perfect sets are uncountable.

Proof. If a perfect set were countable then let F_n be an enumeration of its points. Then each F_n is closed and their union is complete, so by BCT one must contain an interval, and this must correspond to an isolated point.

□

3.3 Divergence of Fourier series

Suppose f is a 2π-periodic function so that $f(x) = f(x + 2\pi)$ for all x.

Then, we will attempt to write f as a Fourier series so that

$$f(x) = \sum_{k \in \mathbb{Z}} a_k e^{ikx}$$

This basically splits f into "frequency components" and is the discrete analog of a Fourier transform.

I will now show a heuristic argument that gives us a formula for the a_n and then we will show that this cannot always work and therefore requires extra justification.

Lemma. Let n be an integer, then $\int_0^{2\pi} e^{inx} dx = 2\pi\delta_{0n}$.

Proof. If $n = 0$ the result is clear since we are integrating 1. Otherwise, the antiderivative is $\frac{e^{inx}}{in}$ which is equal when evaluated at the 2 x values.

□

First, assume $f(x) = \sum a_k e^{ikx}$. Now suspend your disbelief that we can do stuff with infinite sums and integrals that is sufficiently intuitive, then we have

$$a_m = \frac{1}{2\pi} \sum_k a_k (2\pi\delta_{mk}) = \frac{1}{2\pi} \sum_k a_k \int_0^{2\pi} e^{i(k-m)x} dx = \frac{1}{2\pi} \int_0^{2\pi} \sum_k a_k e^{i(k-m)x} dx = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-imx} dx$$

This suggests a formula for a_m, that $a_m = \hat{f}(-m)$ as defined in the probability and measure course. We say the Fourier Series for f is given by $f(x) \sim \sum a_k e^{ikx}$. Similar to Taylor series, we don't know when this converges or equals the function and will need to develop conditions for it.

Theorem. There exists a continuous 2π-periodic function f whose Fourier series diverges at $x = 0$

Proof. Define the partial sums as follows:

$$S_N f(0) = \sum_{k=-N}^{N} \hat{f}(k)$$

Then note $f \mapsto S_N f(0)$ is a linear operator which we will call ϕ_N. It is defined on the space X of 2π-periodic continuous functions which is isomorphic to the space $\{f \in C : f(0) = f(2\pi)\}$. With the sup norm, this is a closed subspace of $C[0, 2\pi]$ so it is a Banach space.

Assume for a contradiction that for all $f \in X$ we have $\sup_N \|\phi_N f\| < \infty$.

By the Banach-Steinhaus uniform boundedness principle, it is enough to show that $\|\phi_N\|$ is unbounded to reach a contradiction.

Define $D_N(x) = \sum_{k=-N}^{N} e^{ikx}$. Then we have

$$\phi_N f = \sum_{k=-N}^{N} \hat{f}(k) = \frac{1}{2\pi} \sum_{k=-N}^{N} \int_0^{2\pi} f(x) e^{-ikx} dx = \frac{1}{2\pi} \int_0^{2\pi} f(x) D_N(x) dx$$

Now we have

$$\|\phi_N f\| \leq \frac{1}{2\pi} \left(\sup_x |f(x)| \right) \int_0^{2\pi} |D_N(x)| dx$$

This means that $\|\phi_N\| \leq \frac{1}{2\pi} \int_0^{2\pi} |D_N(x)| dx$. To show that $\|\phi_N\| = \frac{1}{2\pi} \int_0^{2\pi} |D_N(x)| dx$ we take $f(x) = e^{-i * \arg(D_N(x))}$, then $\|f\|_\infty = 1$ and

$$\|\phi_N f\| = \frac{1}{2\pi} \int_0^{2\pi} |D_N(x)| dx$$

Now we just need to show that $\int_0^{2\pi} |D_N(x)| dx \to \infty$.

We have

$$D_N(x) = e^{-iNx} \left(1 + e^{ix} + ... + e^{2Nix}\right) = \frac{e^{-iNx} \left(e^{(2N+1)ix} - 1\right)}{e^{ix} - 1}$$
$$= \frac{e^{\left(N+\frac{1}{2}\right)ix} - e^{-\left(N+\frac{1}{2}\right)ix}}{e^{ix/2} - e^{-ix/2}} = \frac{\sin\left(\left(N + \frac{1}{2}\right)x\right)}{\sin\left(\frac{1}{2}x\right)}$$

Note that $|D_N(x)|$ gets large when for an integer r, $\left(N + \frac{1}{2}\right)x = \pi\left(r + \frac{1}{2}\right)$. This makes $x_r = \frac{\pi\left(r+\frac{1}{2}\right)}{N+\frac{1}{2}}$ and we will consider r between 1 and N. In what follows I will use some very crude but sufficient bounds (like $\pi < 4$).

We consider the contribution to the integral from points within a width $\frac{1}{10N}$ around these peaks. In this safe region,

$$\left|\sin\left(\left(N + \frac{1}{2}\right)x\right)\right| \geq \frac{1}{2}$$

but

$$\sin\left(\frac{1}{2}x\right) \leq \frac{1}{2}x \leq \frac{1}{2}x_r + \frac{1}{40N} = \frac{\pi\left(r + \frac{1}{2}\right)}{2\left(N + \frac{1}{2}\right)} + \frac{1}{40N} < \frac{8r}{2N} + \frac{1}{40N} = \frac{5r}{N}$$

Therefore in a width of $\frac{1}{10N}$ around the r'th peak the thing is at least $\frac{N}{10r}$ so we contribute at least $\frac{1}{100r}$ to the integral. This means that

$$\int_0^{2\pi} |D_N(x)| dx \geq \sum_{r=1}^{N} \frac{1}{100r} \to \infty$$

□

4 The space $C(X)$

4.1 Basic properties

Let X be a topological space, then $C(X)$ is the space of continuous $\mathbb{R}$-valued functions on X.

Definition. A topological space is locally compact if for every point x there exists an open set U and a compact set K such that $x \in U \subseteq K$.

Recall that a Hausdorff topological space is one where any two distinct points have disjoint open intervals around them.

Recall that if X is a compact Hausdorff space and $A \subseteq X$ then A is compact if and only if A is closed.

Definition. A topological space X is said to be normal if for every pair $A, B \subseteq X$ of disjoint closed sets there exist disjoint open sets U, V with $A \subseteq U$ and $B \subseteq V$.

Proposition. Suppose X is a compact Hausdorff space, then X is normal.

Proof. Suppose $A, B \subseteq X$ are disjoint closed sets. For every $a \in A$ and $b \in B$ by hausdorffness there are disjoint open sets $U_{a,b}$ and $V_{a,b}$ containing a and b respectively.

Now fix $a \in A$. The sets $V_{a,b}, b \in B$ form an open cover of B, and since B is compact (since closed sets are compact in a compact Hausdorff space) there is a finite subcover. The union of this finite subcover, $V_{a,b_1} \cup V_{a,b_2} \cup ... \cup V_{a,b_n}$, is something we will call V_a, and we will write U_a for the intersection $U_{a,b_1} \cap U_{a,b_2} \cap ... \cap U_{a,b_n}$ which is open.

Then U_a and V_a are open and disjoint with $a \in U_a$ and $B \subseteq V_a$. But now the U_a form an open cover of A and isnce A is compact there is a finite subcover $U_{a_1}, U_{a_2}, ..., U_{a_m}$. Take U to be the union of this subcover and V to be the intersection $V_{a_1} \cap V_{a_2} \cap ... \cap V_{a_m}$ to get the disjoint open sets we want.

□

Lemma. Let X be a topological space. Then X is normal if and only if it has the property that if A is closed and W is open and $A \subseteq W$ then there is an open set U with $A \subseteq U \subseteq \overline{U} \subseteq W$.

Proof. Suppose X has the property above and let $A, B \subseteq X$ be disjoint closed sets. Then take $W = X \setminus B$, then W is open and $A \subseteq W$. Find the U generated by the property above and let $V = X \setminus \overline{U}$, then $B \subseteq V$ so the space is normal.

Conversely, if the space is normal, then we prove it has the property above as follows: Let A be a closed subset of X and let W be an open subset of X such that $A \subseteq W$. Now let $B = X \setminus W$, then this is closed, and there exist disjoint open sets around A, B by normality, call these U, V respectively, then we have $A \subseteq U \subseteq \overline{U} \subseteq X \setminus V \subseteq W$.

□

Theorem. (Urysohn's Lemma). Let X be a compact Hausdorff space, and let $A, B \subseteq X$ be disjoint closed sets, then there exists a continuous function $f : X \to [0,1]$ with $f|_A = 0$ and $f|_B = 1$.

Remark. If X is a metric space take $f(x) = \frac{\text{dist}(x,A)}{\text{dist}(x,A)+\text{dist}(x,B)}$.

Proof. Set $W = X \setminus B$. Then this is an open set containing A. By the lemma above there exists open sets U_0 and U_1 such that

$$A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1 \subseteq \overline{U_1} \subseteq W$$

We can repeatedly apply the lemma above to find an open set $U_{\frac{1}{2}}$ such that

$$A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_{\frac{1}{2}} \subseteq \overline{U_{\frac{1}{2}}} \subseteq U_1 \subseteq \overline{U_1} \subseteq W$$

Next we can find open sets $U_{\frac{1}{4}}$ and $U_{\frac{3}{4}}$ such that

$$A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_{\frac{1}{4}} \subseteq \overline{U_{\frac{1}{4}}} \subseteq U_{\frac{1}{2}} \subseteq \overline{U_{\frac{1}{2}}} \subseteq U_{\frac{3}{4}} \subseteq \overline{U_{\frac{3}{4}}} \subseteq U_1 \subseteq \overline{U_1} \subseteq W$$

We can repeat this process to do this for all dyadic rationals (numbers of the form $\frac{a}{2^b}$ with a, b integers) between 0.

We end up with open sets U_r for each dyadic rational r with $0 \leq r \leq 1$ with the property that $r < s$ implies $\overline{U_r} \subseteq U_s$.

For $x \in U_1$ define $f(x) = \inf \{r : x \in U_r\}$ and define $f(x) = 1$ otherwise. Then this f has the properties we want, as we will prove the non-obvious one which is that it is not continuous. To prove this, we will first prove that if $\alpha \in (0,1)$ then $f^{-1}([0,\alpha])$ and $f^{-1}([\alpha,1])$ are closed which implies that pre-images of open intervals and thus open sets are open, as if $\alpha = 1$ or $\alpha = 0$ then we take an intersection of a sequence of α tending to those values respectively which will therefore also be closed. To prove that claim, we will prove that (1) $f^{-1}([0,\alpha]) = \cap_{r>\alpha}\overline{U_r}$ and that (2) $f^{-1}([\alpha,1]) = X \setminus \cup_{r<\alpha}U_r$, or equivalently $f^{-1}([0,\alpha)) = \cup_{r<\alpha}U_r$.

To prove (1) note that if $x \in f^{-1}([0,\alpha])$ then $f(x) \leq \alpha$ so $x \in U_r$ for all $r > \alpha$ and therefore $x \in \cap_{r>\alpha}U_r$. Conversely, if $x \in \cap_{r>\alpha}U_r$ then $x \in \overline{U_r}$ for all $r > \alpha$ so $x \in U_s$ for all $s > r > \alpha$ and thus all $s > \alpha$. Therefore $f(x) \leq \alpha$ so $x \in f^{-1}([0,\alpha])$.

To prove (2) note that

$$x \in f^{-1}([0,\alpha)) \Longleftrightarrow f(x) < \alpha \Longleftrightarrow x \in U_r \text{ for some } r < \alpha \Longleftrightarrow x \in \cup_{r<\alpha}U_r$$

□

Theorem. (Tietze Extension Theorem) Suppose X is a compact Hausdorff space. Suppose $S \subseteq X$ is closed and that $f : S \to [-1,1]$ is a continuous function, then there is a continuous function $\tilde{f} : X \to [-1,1]$ such that $\tilde{f}|_S = f$.

Proof. First, let $A_0 = \left\{x \in S : f(x) \leq -\frac{1}{3}\right\}$ and $B_0 = \left\{x \in S : f(x) \geq -\frac{1}{3}\right\}$, then these are disjoint closed sets. Also define $f_0 = f$ on S.

By Urysohn's Lemma (rescaled), there is a continuous function $g_0 : X \to \left[-\frac{1}{3}, \frac{1}{3}\right]$ such that $g_0|_{A_0} = -\frac{1}{3}$ and $g_0|_{B_0} = \frac{1}{3}$.

Define $f_1 = f_0 - g_0$ on S. Then note that f_1 is a function $f_1 : S \to \left[-\frac{2}{3}, \frac{2}{3}\right]$.

Now define $A_1 = \left\{x \in S : f_1(x) \leq -\frac{1}{3} * \frac{2}{3}\right\}$ and $B_1 = \left\{x \in S : f_1(x) \leq \frac{1}{3} * \frac{2}{3}\right\}$.

By Urysohn's Lemma (again), there is a continuous function $g_1 : X \to \left[-\frac{2}{9}, \frac{2}{9}\right]$ such that $g_1|_{A_1} = -\frac{2}{9}$ and $g_1|_{B_1} = \frac{2}{9}$. Define $f_2 = f_1 - g_1$

We can continue in this way to obtain functions f_n with $|f_n(x)| \leq \left(\frac{2}{3}\right)^n$ for all $x \in S$ and $A_n = \left\{x \in S : f_n(x) \leq -\frac{1}{3}\left(\frac{2}{3}\right)^n\right\}$ and $B_n = \left\{x \in S : f_n(x) \geq \frac{1}{3}\left(\frac{2}{3}\right)^n\right\}$ and functions $g_n : X \to \left[-\frac{1}{3}\left(\frac{2}{3}\right)^n, \frac{1}{3}\left(\frac{2}{3}\right)^n\right]$ continuous with $g_n|_{A_n} = -\frac{1}{3}\left(\frac{2}{3}\right)^n$ and $g_n|_{B_n} = \frac{1}{3}\left(\frac{2}{3}\right)^n$.

Define $\tilde{f}(x) = g_0(x) + g_1(x) + ...$, then this converges uniformly on X since $\|g_n\| \leq \frac{1}{3}\left(\frac{2}{3}\right)^n$. Therefore the uniform limit is continuous.

Also, the uniform limit is $\tilde{f}$ since when we constructed the f_n's it was in such a way that $f_n = \tilde{f} - g_0 - g_1 - ...$ converges uniformly to 0.

□

4.2 Stone-Weierstrass Theorem

We will generalize the theorem from Analysis that a continuous function on $[a,b]$ can be uniformly approximated by polynomials.

Definition. Let X be a topological space and let $A \subseteq C(X)$ be a set of functions. Then we say that A is an algebra if whenever $f, g \in A$ we have $f + g \in A$ and $fg \in A$ and $\lambda f \in A$ for all $\lambda \in \mathbb{R}$.

Note that if X is compact then any continuous function is bounded by the topological version of the Extreme value theorem. This means that $C(X)$ can be equipped with the sup norm.

Example. On $X = [a, b]$ the set of real-coefficient polynomials is an algebra.

Example. On $X = [a_1, b_1] \times [a_2, b_2] \subset \mathbb{R}^2$ an algebra is given by the collection of finite sums $\sum_{i=1}^{n} g_i(x) h_i(y)$ with g, h being continuous one-variable functions.

Definition. An algebra A is said to separate points if for every $x, y \in X$ distinct and $s, t \in \mathbb{R}$ there is an $f \in A$ such that $f(x) = s$ and $f(y) = t$

Theorem. (Stone-Weierstrass theorem) Let X be a compact Hausdorff topological space. Let $A \subseteq C(X)$ be an algebra of functions that separate points, then $\overline{A} = C(X)$ where $\overline{A}$ is the closure of A with the sup norm.

Proof. **Step 1.** We will prove that if $f, g \in \overline{A}$ then $\max(f, g) \in \overline{A}$ and $\min(f, g) \in \overline{A}$.

To do this we will prove that if $f \in \overline{A}$ then $|f| \in \overline{A}$ as this will imply that $\max(f, g) = g + \frac{1}{2}(f - g + |f - g|) \in \overline{A}$ and therefore that $\min(f, g) = -\max(-f, -g) \in \overline{A}$.

Since f is bounded, we can rescale it in such a way that we assume $|f(x)| \leq 1$ for all x.

Now note that by the polynomial approximation theorem from Analysis, there exists a sequence $(P_n)_{n=1}^{\infty}$ of polynomials on $[-1, 1]$ that converges to $|x|$. By taking this sequence and subtracting the constant term (which tends to 0) from each polynomial, we get a sequence of such polynomials with zero constant term on $[-1, 1]$ that converges to $|x|$ uniformly.

Since $\overline{A}$ is closed under uniform limits, and contains all functions of the form $P_n(f)$ and f is bounded by 1, it follows that it must contain $|f|$.

Step 2. We will now prove that $\overline{A} = C(X)$

We say A is a lattice if it has the property that if $f, g \in A$ then $\max(f, g), \min(f, g) \in A$. Now consider any lattice $A \subseteq C(X)$ that separate points and let $f \in C(X)$ be arbitrary, then we will prove that f can be approximated uniformly by functions from A.

Fix $\varepsilon > 0$. Given any $x, y \in X$, there is a function $f_{x,y} \in A$ such that $f_{x,y}(x) = f(x)$ and $f_{x,y}(y) = f(y)$. Let

$$U_{x,y} = \{z \in X : f_{x,y}(z) < f(z) + \varepsilon\}$$

Then these sets are open and $x, y \in U_{x,y}$

Now for each fixed x if we range over all y then these sets form an open cover so there is a finite subcover by compactness, the subcover can be written as $U_{x,y_1}, U_{x,y_2}, ..., U_{x,y_n}$. Define $f_x = \min\{f_{x,y_1}, ..., f_{x,y_n}\}$, then $f_x \in A$ and $f_x(x) = f(x)$ and $f_x(z) < f(z) + \varepsilon$ for all $z \in X$.

For each x, define $V_x = \{z \in X : f_x(z) > f(z) - \varepsilon\}$. Againm these form an open cover, so we pass to a finite subcover $V_{x_1}, ..., V_{x_n}$ by compactness, and define $\tilde{f} = \max\{f_{x_1}, ..., f_{x_n}\}$. Then by construction $\tilde{f} \in A$ and $f(z) - \varepsilon < \tilde{f}(z) < f(z) + \varepsilon$ for every $z \in X$. Thus, A is indeed dense in $C(X)$ so the result follows.

□

Corollary. (Stone-Weierstrass, complex form) Let X be a compact Hausdorff space. Suppose A is an algebra of complex-valued functions that separate points and has the additional property that if $f \in A$ then the complex conjugate of f, $\bar{f}$, is in A.

Then $\overline{A} = C^{\mathbb{C}}(X)$ where this means the space of complex-valued continuous functions on X, with the sup norm.

Proof. For every function $f \in A$, the real part $\operatorname{Re}(f) = \frac{1}{2}\left(f + \bar{f}\right)$ and imaginary part $\operatorname{Im}(f) = \frac{1}{2i}\left(f - \bar{f}\right)$ are both in A. It follows that the real algebra $A \cap C^{\mathbb{R}}(X)$ separates points, and by the real version of Stone-Weierstrass, A is dense in $C^{\mathbb{R}}(X)$. But then the result follows as $C^{\mathbb{C}}(X) = C^{\mathbb{R}}(X) + iC^{\mathbb{R}}(X)$

□

Corollary. Every continuous complex-valued 2π-periodic function can be uniformly approximated by finite sums of exponentials of the form $\sum_{n=-N}^{N} a_n e^{in\theta}$, but this does not mean the fourier series converges as the a_k's may depend on N.

Proof. Apply the above proof to the algebra of sums of exponentials on the unit circle. Note that separation of points holds by considering functions of the form $a + be^{i\theta}$ and it is closed under conjugates by reversing the order of each sum.

□

Theorem. Let X be a compact Hausdorff space. Then any proper closed subalgebra of $C(X)$ is contained in at least one of either

i) The subalgebras $A_{x,y} = \{f \in C(X) : f(x) = f(y)\}$ for $x \neq y$

ii) The subalgebras $A_x = \{f \in C(X) : f(x) = 0\}$

Proof. Fix x and y. Let A be a proper sub-algebra and define $V_{x,y} = \{(f(x), f(y)) : f \in A\} \subseteq \mathbb{R}^2$. Note that this is a vector space.

Assume for a contradiction that A is not contained in any $A_{x,y}$ or A_x or A_y. Then this implies that $V_{x,y}$ contains points of the form

$$(a_1, b_1), (a_2, b_2), (a_3, b_3) \text{ with } a_1 \neq b_1, a_2 \neq 0, b_3 \neq 0$$

. Therefore, if $V_{x,y} \neq \mathbb{R}^2$, it must be the case that $V_{x,y} = \langle (t, u) \rangle$ with $t \neq u, t \neq 0, u \neq 0$. But also, $V_{x,y}$ contains $\left(t^2, u^2\right)$ which is linearly independent by the conditions on t, u above. Therefore, $V_{x,y} = \mathbb{R}^2$ which implies that A separates points at x, y. Since x, y were arbitrary, it follows from Stone-Weierstrass that $\overline{A} = C(X)$.

Therefore if A was closed and not $C(X)$ it has to be contained in one of those sets otherwise its closure which is just A would be $C(X)$.

□

Corollary. (Alternative formulation of Stone-Weierstrass) Let X be a compact Hausdorff Topological space and let $A \subseteq C(X)$ be an algebra that separates points. Then either A is dense in $C(X)$ (with respect to sup norm topology) or there is $x_0 \in X$ such that $f(x_0) = 0$ for all $f \in A$.

Proof. The assumption is that $A \not\subseteq A_{x,y}$ for all x, y. This means that $\overline{A}$ is either everything or contained in A_x for some x by the above theorem.

□

Sometimes it might be useful to talk about what happens if the underlying space is not compact, for example what if $X = \mathbb{R}$.

Suppose X is locally compact, which $\mathbb{R}$ is. Define

$$C_0(X) = \{f \in C(X) : \{x : |f(x)| \geq \varepsilon\} \text{ is compact for all } \varepsilon > 0\}$$

Proposition. $C_0(X)$ is a closed subalgebra

Proof. Let f_n be a sequence of functions where each $f_n \in C_0(X)$ and suppose $f_n \to f$ uniformly, then we know f is continuous and want to prove it is in C_0.

By uniform convergence there is some integer N such that for all $x \in X$ we have $|f_N(x) - f(x)| < \frac{\varepsilon}{2}$. Then define

$$K_N = \left\{ x \in X : |f_N(x)| \geq \frac{\varepsilon}{2} \right\}$$

This is a compact set. Now let x be an element of our target set K (meaning $|f(x)| \geq \varepsilon$). Then by the reverse triangle inequality we have

$$|f_N(x)| = |f_N(x) - f(x) + f(x)| \geq |f(x)| - |f_N(x) - f(x)|$$

Therefore

$$|f_N(x)| \geq \varepsilon - \frac{\varepsilon}{2} = \frac{\varepsilon}{2}$$

This forces $x \in K_N$. Therefore $K \subseteq K_N$. Thus, K is bounded. K is closed because it is the pre-image of a closed set under a continuous function, thus K is compact.

□

Theorem. (Stone-Weierstrass for locally compact spaces) Let X be a locally compact Hausdorff topological space and suppose $A \subseteq C_0(X)$ is an algebra that separates points, then $\overline{A} = C_0(X)$.

Proof. Consider the "one-point compactification" X' of X. This is the space $X \cup \{\infty\}$ where ∞ is an extra element and the open sets in X' are the sets that are open in X together with sets of the form $(X \setminus K) \cup \{\infty\}$ for some compact set $K \subseteq X$.

For example, in the reals, this looks like Figure 3, where the reals become like the unit sphere which is compact.

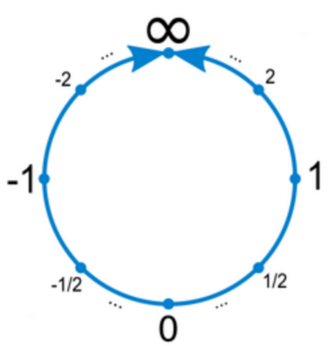

Figure 3

To check that X' becomes Hausdorff, note that this follows from local compactness (it is here that we actually use local compactness). We just need to check that there are disjoint open sets containing x and ∞. If $x \in X$ then there is an open set U with compact closure $\overline{U}$, and thus $V = \overline{U}^C$ is open and contains ∞ and is disjoint from U.

We can check that $C_0(X)$ can be identified with the space

$$\left\{ \tilde{f} \in C(X') : \tilde{f}(\infty) = 0 \right\}$$

where $f \in C_0(X)$ is extended to be 0 at ∞. To verify this claim, we must check that $\tilde{f}$ is continuous. This is because the pre-image of an open set not containing 0 is open and the pre-image of an open interval around 0 is also open.

Under this identification, A can be identified with an algebra $A' \subseteq C(X')$. Let B be the subalgebra of $C(X')$ generated by A' and the constant functions. Then B separates points in $C(X')$ and since there is no point where any function in B is 0, B is dense in $C(X')$.

Note that everything in B is just a constant plus something in A'. To prove this, we need to show that the set of all such functions is an algebra. Indeed, if $f_1, f_2 \in A'$ then

$$(f_1 + c)(f_2 + d) = f_1 f_2 + c f_1 + c f_2 + cd$$

which is cd plus something in A'. The other algebra properties follow similarly.

Now let $f \in C_0(X)$. Then by the work we just did, for every $\varepsilon > 0$ there is some $\tilde{g} \in A'$ and a constant c such that $\tilde{f} = \tilde{g} + c + \eta$ where $\tilde{g} + c \in B$ and $\|\eta\|_\infty < \frac{\varepsilon}{2}$.

Evaluating at ∞ we get $|c| < \frac{\varepsilon}{2}$ because $\tilde{f} = \tilde{g} = 0$. Thus, $\left\|\tilde{f} - \tilde{g}\right\|_\infty < \varepsilon$ and so as required there is some $g \in A$ such that $\|f - g\|_\infty < \varepsilon$.

□

Example. Let $f : [0, \infty) \to \mathbb{R}$ be such that f approaches 0 as $x \to \infty$, then f can be uniformly approximated by functions of the form $\sum_{r=1}^{n} a_r e^{-rx}$.

4.3 Arzela-Ascoli Theorem

Definition. Let $F \subseteq C(X)$. We say F is precompact if its closure $\overline{F}$ is compact.

Example. $\{f : C[0,1] : \|f\|_\infty \leq 1\}$ is not compact because an open cover with no finite subcover is given by open balls around bump functions supported on $[n, n+1]$.

Definition. We say F is uniformly bounded if $\sup_{f \in F} \|f\|_\infty < \infty$

Definition. We say F is equicontinuous if for each $x \in X$ and for all $\varepsilon > 0$ there exists U_x open with $x \in U_x$ and such that $y \in U_x$ implies $|f(y) - f(x)| < \varepsilon$ for all $f \in F$ simultaneously.

Theorem. (Arzela-Ascoli) Let X be a compact Hausdorff topological space and let $F \subseteq C(X)$, then F is precompact if and only if F is uniformly bounded and equicontinuous

We'll prove this shortly, but first we want some remarks.

Note that the $UB + EQ \implies PC$ direction with $X \subset \mathbb{R}^2$ and F countable implies the version of this theorem we proved in level 6.4 which we proved by repeated Bolzano-Weierstrass application.

Note that by compactness, equicontinuity at each point implies that there is a δ that works for every x so we can drop that global condition. The rough idea is that for each x find a δ-ball that works for $\frac{\varepsilon}{4}$, then cover the domain by the corresponding $\frac{\delta}{2}$-balls and take a finite subcover. Then pick the minimum such δ from the finite subcover, call it η, then 2 points within η must be within neighbouring balls so f must differ by at most ε between any points less than η apart.

Recall that F is totally bounded if for all $\varepsilon > 0$ we can cover F by finitely many open balls of radius ε centered at a point in F.

Lemma. Total boundedness of F and $\overline{F}$ are equivalent.

Proof. Suppose first that F has been covered by finitely many balls $B\left(f_i, \frac{\varepsilon}{2}\right)$. If $g \in \overline{F}$ then there is a sequence of functions in F with limit g. At least one of the balls contains infinitely many elements of the sequence since there are finitely many balls, and g is in the closure of that ball, which is contained in $B(f_i, \varepsilon)$

On the other hand, if $\overline{F}$ has been covered by open balls of radius $\frac{\varepsilon}{2}$ about some point f_i then for each i choose $g_i \in F$ with $d(f_i, g_i) < \frac{\varepsilon}{2}$, then the balls $B(g_i, \varepsilon)$ cover F.

□

Recall from Analysis that a metric space is compact if and only if it is complete and totally bounded. Note that $\overline{F}$ is already complete because it is a closed subset of a complete metric space $C(X)$. We will now prove Arzela-Ascoli by showing that Total Boundedness of F which is equivalent to Total Boundedness of $\overline{F}$ which is equivalent to compactness of $\overline{F}$ which is equivalent to precompactness of $\overline{F}$, is equivalent to Uniform Boundedness + Equicontinuity.

Proof. **Step 1.** We will prove that totally bounded implies uniformly bounded.

For any $\varepsilon > 0$ there is some finite collection $\{f_1, f_2, ..., f_n\}$ of functions such that for all $f \in F$ one of the f_i's has $\|f - f_i\|_\infty \leq \varepsilon$.

In particular, $\sup_{f\in F} \|f\|_\infty \leq \max \|f_i\|_\infty + \varepsilon$. Therefore, F is uniformly bounded.

Step 2. We will prove that totally bounded implies equicontinuous.

Work with the same set $\{f_1, f_2, ..., f_n\}$ as above. At each x, there is an open set U containing x such that if $y \in U$ then $|f_i(x) - f_i(y)| \leq \frac{\varepsilon}{3}$ for each $i = 1, 2, ..., n$. Now if $f \in F$ is arbitrary, then let $y \in U$ and choose i such that $\|f - f_i\|_\infty \leq \frac{\varepsilon}{3}$. Then

$$|f(y) - f(x)| \leq |f(y) - f_i(y)| + |f_i(y) - f_i(x)| + |f_i(x) - f(x)| \leq 3\frac{\varepsilon}{3} = \varepsilon$$

This implies that F is equicontinuous since there is a U that works for every x.

Step 3. We will prove that equicontinuous and uniformly bounded implies totally bounded.

Let $\varepsilon > 0$. For each $x \in X$ equicontinuity implies that there is an open set U_x with $x \in U_x$ such that $|f(y) - f(x)| < \frac{\varepsilon}{3}$ whenever $f \in F$ and $y \in U_x$. Since X is compact we may pass to a finite subcover $U_{x_1}, U_{x_2}, ..., U_{x_n}$.

Now consider the set of vectors in $\mathbb{R}^n$ given by $(f(x_1), f(x_2), ..., f(x_n))$ for $f \in F$. Since F is uniformly bounded these all are in some box $[-M, M]^n$.

This means that there is a finite set of $\frac{\varepsilon}{3}$-balls that covers this set of vectors. This corresponds to a collection of functions $f_1, f_2, ..., f_k \in F$ such that for each $f \in F$ there is some i such that for each $j = 1, 2, ..., n$ we have $|f(x_j) - f_i(x_j)| \leq \frac{\varepsilon}{3}$.

If $y \in U_{x_j}$ then by equicontinuity we have

$$|f(y) - f_i(y)| \leq |f(y) - f(x_j)| + |f(x_j) - f_i(x_j)| + |f_i(x_j) - f_i(y)| < 3\frac{\varepsilon}{3} = \varepsilon$$

Since the U_{x_j} cover X this is true for all $y \in X$.

Therefore, $\|f - f_i\|_\infty < \varepsilon$.

We have shown now that the balls of radius ε about $f_1, f_2, ..., f_k$ in the sup norm cover F. Thus, F is totally bounded because ε was arbitrary.

□

Example. This theorem implies that for fixed c_1, c_2 the set of functions that are Lipschitz continuous with c_1 as the Lipschitz constant and bounded by c_2, is precompact.

Recall from level 6.4 that Arzela-Ascoli can be used to prove the Peano Existence theorem for ODEs which can in turn be used to prove the 2D implicit function theorem which was the whole reason level 6.4 exists in the first place.

5 Weak topologies on normed spaces

If X is a normed space then we always have the usual topology. However, there is another topology we can define.

Let f be a family of functions going from X to another topological space. This could be a different topological space for each f, so we will denote it Y_f. We will then define a topology as follows:

$$S = \left\{f^{-1}(U) : U \in T_{Y_f}, f \in F\right\}$$

$$T_F = \bigcap T \text{ over topologies T on X with } S \subseteq T$$

In other words this is the coarsest topology that makes every $f \in F$ continuous. It is the set of arbitrary unions of finite intersections of sets of the form $f^{-1}(U)$ where $U \in T_{Y_f}$ and $f \in F$.

Definition. F is said to separate points of X if for all $x, y \in X$ there exists $f \in F$ such that $f(x) \neq f(y)$

Proposition. Let F be a family of maps $f : X \to Y_f$ and suppose that Y_f is Hausdorff for all $f \in F$, and F separates points of X. Then (X, T_F) is Hausdorff.

Proof. Let $x, y \in X$, $x \neq y$. Then there exists $f \in F$ with $f(x) \neq f(y)$. By Hausdorffness there exist open $W_1, W_2 \in T_{Y_f}$ such that $f(x) \in W_1$ and $f(y) \in W_2$ and $W_1 \cap W_2 = \emptyset$.

Then $x \in f^{-1}(W_1) \in T_F$ and $y \in f^{-1}(W_2) \in F$ and $f^{-1}(W_1) \cap f^{-1}(W_2) = \emptyset$ so (X, T_F) is Hausdorff.

□

Lemma. Let $f_1, f_2, ..., f_n$ be linear functionals on X and let $N = \{x \in X : f_i(x) = 0 \text{ for all } i\}$. Then $f(x) = 0$ for all $x \in N$ if and only if there exist $a_1, a_2, ..., a_n$ such that $f = \sum a_i f_i$

Proof. It is clear that if $f = \sum a_i f_i$ then $f = 0$ for all $x \in N$.

If $f = 0$ for all $x \in N$ then let $\mathbb{F}$ be the underlying field and define $T : X \to \mathbb{F}^n$ by

$$T(x) = (f_1(x), f_2(x), ..., f_n(x))$$

Note that $\ker(T) = N \subseteq \ker(f)$. Therefore the value of $f(x)$ depends only on the value of $T(x)$ since if $T(x) = T(y)$ then $T(x - y) = 0$ so $x - y \in \ker(f)$ so $f(x) = f(y)$. This means there is a function $g : \text{im}(T) \to \mathbb{F}$ such that

$$g(T(x)) = f(x)$$

Since $\text{im}(T) \subseteq \mathbb{F}^n$ we can extend g to a linear funcitonal $G : \mathbb{F}^n \to \mathbb{F}$. Then $G(y_1, y_2, ..., y_n) = \sum_{i=1}^n a_i y_i$. Thus, we have that $f(x) = G(f_1(x), f_2(x), ..., f_n(x)) = \sum_{i=1}^n a_i f_i(x)$.

□

Definition. A topological vector space X is a vector space over a field $\mathbb{K}$ that is also a topological space (Here it will be $\mathbb{R}$ or $\mathbb{C}$ with the usual topology) that has its own topology such that the functions of addition $\cdot + \cdot : X \times X \to X$ and scalar multiplication $\mathbb{K} \times X \to X$ are both continuous functions with respect to the product topologies.

Definition. In a topological vector space, its dual is the set of continuous linear functionals on that vector space. If the topology is not induced by an underlying norm, this is not necessarily the set of bounded linear functionals.

Definition. A neighbourhood basis for a point x in a topological space is a family of open sets $\mathcal{V}$ such that for every open set $U \supseteq X$ there exists $V \in \mathcal{V}$ such that $V \subseteq U$.

Definition. A locally convex space is a topological vector space with a neighbourhood basis consisting of convex sets.

Theorem. Let X be a vector space and let F be a vector space of linear functionals X to $\mathbb{R}$ or $\mathbb{C}$ with their usual topologies that separates points of X. Then (X, T_F) is locally convex and the dual space, $(X, T_F)^* = F$.

Proof. Since F separates the points of X we know that (X, T_F) is Hausdorff. We first check the existence of a basis consisting of convex sets. Recall that T_F is the set of arbitrary unions of finite intersections of sets of the form $f^{-1}(U)$ where $U \in T_{Y_f}$ and $f \in F$. Therefore a neighbourhood basis about 0 for is sets of the form

$$V = \bigcap_{i=1}^{n} f_i^{-1}(B_\varepsilon(0))$$

for some $f_i \in F$ and $\varepsilon > 0$.

V is always convex because if $x, y \in V$ and $\alpha \in [0, 1]$then by linearity we have $\alpha x + (1 - \alpha) y \in V$ because

$$|f_i(\alpha x + (1 - \alpha) y)| = |\alpha f_i(x) + (1 - \alpha) f_i(y)| \leq \alpha |f_i(x)| + (1 - \alpha) |f_i(y)| < \varepsilon$$

To conclude that (X, T_F) is locally convex we need to verify that it is actually a topological vector space.

If V is a 0-neighbourhood as before, then the sum $\frac{1}{2}V + \frac{1}{2}V = V$ ($+$ meaning sum of any 2 things in $\frac{1}{2}V$), since clearly $V \subseteq \frac{1}{2}V + \frac{1}{2}V$, and by convexity if $a, b \in \frac{1}{2}V$ so is $\frac{a+b}{2}$ so $a + b \in V$.

Therefore, we have

$$\left(x + \frac{1}{2}V\right) + \left(y + \frac{1}{2}V\right) = x + y + V$$

Therefore, if $z = x + y$ and $z \in U \in T_F$ then there is a V as above such that $z + V \subseteq U$ so $\left(x + \frac{1}{2}V\right) \times \left(y + \frac{1}{2}V\right) \subseteq +^{-1}(x + y + V)$. Since this is true about any z, it follows that $+$ is continuous.

If $z = \lambda x$ and $z \in U \in T_F$ then there is a V as above such that $z + V \subseteq U$. If V is the pre-image of $B_\varepsilon(0)$ under all the functions, then set $\delta = \min\left(1, \frac{\varepsilon}{2M+1}\right)$ where $M = \max_{i \leq 1 \leq n} |f_i(x)|$, and set $V' = \frac{V}{2(|\lambda|+1)}$. Then consider $B_\delta(\lambda) \times (x + V')$. To show that we have a locally convex space, it remains to show that this maps into U. We have, for $a \in B_\delta(\lambda)$ and $y \in x + V'$, that for some $t \in V'$

$$ay - \lambda x = a(x + t) - \lambda x = at + (a - \lambda)x$$

We want to show that this is in V. Apply any f_i from the finite intersection corresponding to V and use the triangle inequality to get

$$|f_i(at + (a - \lambda)x)| \leq |a| |f_i(t)| + |a - \lambda| |f_i(x)|$$

Since $a \in B_\delta(\lambda)$ we have $|a| < |\lambda| + 1$ so

$$< (|\lambda| + 1) \left| \frac{\varepsilon}{2(|\lambda| + 1)} \right| + \delta M = \varepsilon$$

by definition of δ, M and the fact $t \in V'$ and definition of V'. This proves scalar multiplication is continuous so we have a locally convex space.

Note that $F \subseteq (X, T_F)^*$ because F is continuous by definition of the T_F topology.

To show that $(X, T_F)^* \subseteq F$, take $f \in (X, T_F)^*$ and pick an element there exists V in the neighbourhood basis such that $V \subseteq f^{-1}(\{a : |a| < 1\})$ since f is continuous. Write

$$V = \{x \in X : |f_i(x)| < \varepsilon \text{ for all } i = 1, 2, ..., n\}$$

for appropriate $f_1, ..., f_n \in F$.

Let $x_0 \in X$ such that $f_i(x_0) = 0$ for all i. Then $x_0 \in V$ and $ax_0 \in V$ for all a. Therefore $|f(ax_0)| < \varepsilon$ for all a so we must have $f(x_0) = 0$. By the previous lemma it follows that $f = \sum a_i f_i \in F$.

$\square$

Definition. Let X be a normed space and let X^* be the dual space, then the topology on X generated by $F = X^*$ is known as the weak topology on X and is denoted by T_w.

Note that this is Hausdorff by an earlier proposition since X^* separates points by a corollary of Hahn-Banach from earlier.

$U \subseteq X$ is open in the weak topology ('w-open') if and only if for every $x \in U$ there are $f_1, ..., f_n \in X^*$ and $\varepsilon_1, ..., \varepsilon_n > 0$ such that $\{y \in X : \forall i \, |f_i(x) - f_i(y)| < \varepsilon_i\} \subseteq U$. By replacing f_i with $\frac{f_i}{\varepsilon_i}$ we can replace ε_i with 1.

Recall that $\phi_X : X \to X^{**}$ is a function defined defined such that for $f \in X^*$ we have $\phi_X(x)(f) = f(x)$. Recall that is linear and isometric and it may or may not be surjective.

Definition. On X^* the weak-star topology, denoted T_w^* is generated by $F = \{\phi_X(x) : x \in X\}$

$F \subseteq X^*$ is open in the weak-star topology ('w^*-open') if and only if for every $f \in F$ there are $x_1, ..., x_n \in X$ and $\varepsilon_1, ..., \varepsilon_n > 0$ such that

$$\{g \in X^* : \forall i \, |\phi_X(x_i)(f) - \phi_X(x_i)(g)| < \varepsilon_i\} = \{g \in X^* : \forall i \, |f(x_i) - g(x_i)| < \varepsilon_i\} \subseteq F$$

As before we can assume $\varepsilon_i = 1$.

On X, $T_w \subseteq T_{\|\cdot\|}$ (Since we are taking the intersection of all topologies that make maps that are bounded continuous, and this intersection must be contained within one example of such a topology). However, $(X, T_w)^* = \left(X, T_{\|\cdot\|}\right)^*$. On X^*, we have $T_w^* \subseteq T_w \subseteq T_{\|\cdot\|}$, with $T_w^* = T_w$ if X is reflexive.

Lemma. Let X be a normed space. Then $T_w = T_{\|\cdot\|}$ if and only if X is finite dimensional.

Proof. Assume $X = \mathbb{R}^n$ or $X = \mathbb{C}^n$ with norm the norm $\|x\| = \max_i |x_i|$ where $x = (x_1, x_2, ..., x_n)$ (since norms on finite dimensional spaces are equivalent).

Let $f_i(x) = x_i$. For every $\varepsilon > 0$,

$$B_\varepsilon(0) = \{x \in X : |f_i(x)| < \varepsilon \text{ for all } i\} \in T_w$$

Thus, $T_{\|\cdot\|} \subseteq T_w$ so $T_{\|\cdot\|} = T_w$.

Now assume W is a w-open neighbourhood of 0. Then there exist $f_1, f_2, ..., f_n \in X^*$ such that $\{x \in X : |f_i(x)| < 1 \forall i\} \subseteq W$. So $N = \{x \in X : |f_i(x)| = 0 \forall i\} \subseteq W$.

Assume for contradiction $\dim(X) = \infty$ and $T_w = T_{\|\cdot\|}$ then let F be a map $X \to \mathbb{F}^n$ ($\mathbb{F}$ is the underlying field) such that $f(x) = (f_1(x), f_2(x), ..., f_n(x))$. By the first isomorphism theorem from Groups (and therefore Vector spaces which have a group structure with addition), $X/N \cong \operatorname{im}(f) \subseteq \mathbb{F}^n$. Therefore, if N were finite-dimensional, then the Rank-Nullity theorem would imply X is also finite dimensional, therefore $\dim(X) = \infty \implies \dim(N) = \infty$. This means that every w-open neighbourhood of 0 contains an infinite dimensional linear space. Then $B_\varepsilon(0)$, which is open and thus weakly open, is non-zero and contains a vector space, and is therefore unbounded, but it is bounded as it is an epsilon-ball. This is a contradiction.

□

Definition. Let X be a normed space. A sequence (x_n) in X is said to converge weakly ('w-convergent') to $x \in X$ if it converges in the weak topology. We write $x_n \rightharpoonup x$. Recall that this means that x_n eventually intersects all w-open neighbourhoods of x and stays in said open neighbourhoods.

Lemma. Let (x_n) be a sequence in a normed space X. then $x_n \rightharpoonup x$ if and only if for all $f \in X^*$ we have $f(x_n) \to f(x)$

Proof. Assume $x_n \rightharpoonup x$ and $f \in X^*$. Fix $\varepsilon > 0$. Then $V = \{x \in X : |f(x)| < \varepsilon\} \in T_w$. Therefore there exists $N > 0$ such that $x_n \in x + V$ for all $n > N$. Then $|f(x_n - x)| = |f(x_n) - f(x)| < \varepsilon$ for all $n > N$ so $f(x_n) \to f(x)$.

Conversely, suppose $\forall f \in X^* : f(x_n) \to f(x)$ holds.

It suffices to show that if $V = \{x \in X : |f_i(x)| < 1 \text{ for } i = 1, ..., k\}$ then $x_n \in x + V$ for all n sufficiently large.

Then since $f_i(x_n) \to f_i(x)$ for $i = 1, 2, ..., k$, they each are eventually within 1 of $f_i(x)$ for n large enough, say $n > n_i$. Then $x_n \in x + V$ for $n > \max\{n_1, n_2, ..., n_k\}$.

□

Remark. Note that continuity is equivalent to sequential continuity in metric spaces but not in general topological spaces.

Lemma. Let $x_n \rightharpoonup x$. Then x_n is bounded and $\|x\| \leq \liminf_{n\to\infty} \|x_n\|$.

Proof. For each $f \in X^*$, $f(x_n) \to f(x)$ by the previous lemma so there exists c_f such that for all $n \geq 1$ we have $c_f \geq |f(x_n)| = |\phi_X(x_n)(f)|$. In other words, we have a family of functionals on X^* given by $F = \{\phi_X(x_n) : n \geq 1\}$ such that for all $f \in X^*$ there exists $c_f > 0$ with $|T(f)| \leq c_f$ for all $T \in F$. By the Banach-Steinhaus uniform boundedness principle, there exists $c > 0$ such that $\|T\| < c$ for all $T \in F$. Since ϕ_X is an isometry, this implies that $\|x_n\| = \|\phi_X(x_n)\| \leq c$ for all $n \geq 1$ so x_n is bounded.

By a consequence of the Hahn-Banach theorem from earlier, there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x) = \|x\|$. Therefore,

$$\|x\| = f(x) = \lim_{n\to\infty} f(x_n) \leq \liminf_{n\to\infty} \|f\| \|x_n\| = \liminf_{n\to\infty} \|x_n\|$$

□

Definition. A subset $S \subseteq X^*$ is a fundamental subset of X^* if the closure of the span of S is X^* i.e. $\overline{\langle S \rangle} = X^*$.

Lemma. Let (x_n) be a bounded sequence in X and let $S \subseteq X^*$ be a fundamental subset. Then $x_n \rightharpoonup x$ if and only if $f(x_n) \to f(x)$ for all $f \in S$.

Proof. As $S \subseteq X^*$ we know that the direction that $x_n \rightharpoonup x$ implies $f(x_n) \to f(x)$ for all $f \in S$ follows from a previous lemma.

Let $f \in X^*$ and $\varepsilon > 0$. Then there are $f_1, ..., f_m \in S$ and $a_1, ..., a_m$ such that $\|f - \sum a_i f_i\| < \varepsilon$. We have

$$\begin{aligned} |f(x_n) - f(x)| &\leq \left| f(x_n) - \sum a_i f_i(x_n) \right| + \left| \sum a_i f_i(x_n) - \sum a_i f_i(x) \right| + \left| \sum a_i f_i(x) - f(x) \right| \\ &\leq \left\| f - \sum a_i f_i \right\| (\|x_n\| + \|x\|) + \sum |a_i| \, |f_i(x_n) - f_i(x)| \end{aligned}$$

The first term is bounded by a constant times ε and the second term goes to 0 because $f_1, f_2, ..., f_m \in S$.

□

Example. Let $X = l_p$ with $1 < p < \infty$ so that $X^* = l_q$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then $x_n \rightharpoonup x$ if and only if x_n is bounded and $x_n^{(i)} \to x^{(i)}$ for each $i \in \mathbb{N}$.

This is by the above lemma applied to the fundamental subset $S = \{f_i = (0, ..., 0, 1, 0, ...) : i \in \mathbb{N}\}$ where the 1 is in the i'th position. Then the sequence has to be bounded to converge weakly and so if it converges weakly then the above lemma implies the result.

This means that $x_n = (0, ..., 0, 1, 0, ...)$ converges weakly to 0 but not to 0 normally. The point is $f(x_n) \to 0$ for every valid functional f.

Proposition. If $x \in l_1$ then $x_n \rightharpoonup x$ implies $x_n \to x$

In other words, if we have a sequence of uniformly bounded sequences that make absolutely convergent sums and the pointwise limit of these sequences is an absolutely convergent sum, one may swap the sum with the limit.

Remark. If all x_n were known to be bounded above in absolute value by a sum that absolutely converged, we could apply the dominated convergence theorem with the counting measure, but sadly that need not be the case.

Proof. We know $x_n \rightharpoonup x$ implies $f(x_n) \to f(x)$ for all f. Since $l_1^* = l_\infty$ we have that f contains all functionals of the form $(0, ..., 0, 1, 0, ...)$ so it follows that $x_n \rightharpoonup x$ implies the sequences converge pointwise to x.

The idea is that if x_n does not converge to 0 in the norm topology but converges pointwise we will find a functional f such that $f(x_n) \nrightarrow f(x)$. Without loss of generality assume $x = 0$.

If $x_n \nrightarrow 0$ in the norm topology then there exists $\varepsilon > 0$ and a subsequence which we will call x_n such that for all n we have $\|x_n\|_1 \geq \varepsilon$.

Let $n_1 = 1$. Since $\|x_{n_1}\|_1 = \sum_{k=1}^{\infty} \left| x_{n_1}^{(k)} \right|$ is finite, there exists m_1 such that

$$\sum_{k=m_1+1}^{\infty} \left| x_{n_1}^{(k)} \right| < \frac{\varepsilon}{5}$$

Now suppose we have chosen n_{j-1} and m_{j-1}. Then since $x_n \to 0$ coordinate-wise we can find n_j such that

$$\sum_{k=1}^{m_{j-1}} \left| x_{n_j}^{(k)} \right| < \frac{\varepsilon}{5}$$

Because $\left\| x_{n_j} \right\|_1$ is also finite there exists $m_j > m_{j-1}$ such that

$$\sum_{k=m_j+1}^{\infty} \left| x_{n_j}^{(k)} \right| < \frac{\varepsilon}{5}$$

Let $H_j = \{m_{j-1}+1, m_{j-1}+2, ..., m_j\}$. Then

$$\sum_{k \in H_j} \left| x_{n_j}^{(k)} \right| \geq \left\| x_{n_j} \right\|_1 - \frac{2\varepsilon}{5} \geq \frac{3\varepsilon}{5}$$

Now define

$$y_k = \begin{cases} 1 & x_{n_j} \geq 0, \exists k : k \in H_j \\ -1 & x_{n_j} < 0, \exists k : k \in H_j \\ 0 & \text{otherwise} \end{cases}$$

Then $y \in l_\infty$ so y defines a valid bounded linear functional on l_1.

Since $x_{n_j} \rightharpoonup 0$ we must have $y\left(x_{n_j}\right) \to 0$. We also have

$$y\left(x_{n_j}\right) = \sum_{k=1}^{\infty} y_k x_{n_j}^{(k)} = \sum_{k \in H_j} \left| x_{n_j}^{(k)} \right| - \sum_{k \notin H_j} y_k x_{n_j}^{(k)}$$

Therefore by the reverse triangle inequality we have

$$\left| y\left(x_{n_j}\right) \right| \geq \sum_{k \in H_j} \left| x_{n_j}^{(k)} \right| - \sum_{k \notin H_j} \left| x_{n_j}^{(k)} \right|$$

$$\left| y\left(x_{n_j}\right) \right| \geq \frac{3\varepsilon}{5} - \frac{2\varepsilon}{5} = \frac{\varepsilon}{5}$$

so it cannot tend to 0 so we have a contradiction.

□

Note that in l_1, although strong and weak convergence are equivalent, $T_w \neq T_{\|\cdot\|}$ because the space is not finite dimensional.

Definition. A system $\mathcal{F}$ of subsets of a set S is said to be of finite character if for $A \subseteq S$, $A \in \mathcal{F}$ if and only if every finite subset of A is in $\mathcal{F}$.

Lemma. (Tukey's Lemma) Let $\mathcal{F}$ be a set system of finite character and let $F \in \mathcal{F}$. Then $\mathcal{F}$ has a maximal element (with respect to inclusion) containing F.

Proof. Let $F_0 = \{A \in \mathcal{F} : F \subseteq A\}$, so everything in $\mathcal{F}$ containing F. If $C \subseteq F_0$ is a chain then its union, D, is in $\mathcal{F}$ because its finite subsets are in $\mathcal{F}$. Also, $D \in F_0$ since it contains F and it is an upper bound for the chain C. By Zorn's lemma, F_0 has a maximal element which is also a maximal element for $\mathcal{F}$ that contains F.

□

Definition. A family of sets is said to have the finite intersection property if every finite intersection of sets in the family has non-empty intersection.

Proposition. Let X be a topological space. Then X is compact if and only if for every system F of closed subsets of X with the finite intersection property, $\bigcap_{A \in \mathcal{F}} A$ is non-empty.

Proof. X compact (meaning every open cover has a finite subcover)

$\iff$

For every family of closed sets with empty intersection there is a finite subfamily with empty intersection (since complements of open sets are the closed sets)

$\iff$

If no finite subfamily has empty intersection neither does the entire family.

This is the contrapositive of the statement above that is claimed to be equivalent to compactness, so done.

□

Theorem. (Tychonoff's Theorem) The product of a collection of compact spaces is compact with respect to the product topology.

Note that this is easy to prove for finite products by induction and using basic properties of topological spaces, but we will prove it for more general products.

Proof. Let $\{X_\gamma : \gamma \in \Gamma\}$ be a collection of compact spaces and let $X = \prod X_\gamma$. Let $\mathcal{A}$ be a system of subsets of X with the finite intersection property. To show that X is compact we want to show that $\bigcap_{A \in \mathcal{A}} \overline{A} = \emptyset$.

Let $\mathcal{F}$ be the collection of all systems of subsets of X which have the finite intersection property. By definition, $\mathcal{F}$ is of finite character so Tukey's Lemma guarantees the existence of a maximal system $\mathcal{B}$ of subsets of X having the finite intersection property such that $\mathcal{A} \subseteq \mathcal{B}$. Since $\bigcap_{B \in \mathcal{B}} \overline{B} \subseteq \bigcap_{A \in \mathcal{A}} \overline{A}$ it is enough to show that $\bigcap_{B \in \mathcal{B}} \overline{B} = \emptyset$.

Because $\mathcal{B}$ is maximal we know that if $B_1, B_2, ..., B_n \in \mathcal{B}$ then $B_1 \cap B_2 \cap ... \cap B_n \in \mathcal{B}$, for if it weren't in $\mathcal{B}$ we could just add it and still have the finite intersection property, contradicting maximality.

Observation: If $C \subseteq X$ such that $C \cap B$ is non-empty for all $B \in \mathcal{B}$ then $C \in \mathcal{B}$, by similar reasoning.

Now write $x \in X$ as $(x_\gamma)_{\gamma \in \Gamma}$ for $x_\gamma \in X_\gamma$. For each $\gamma \in \Gamma$ let π_γ be the projection $x \mapsto x_\gamma$. Then the system $\{\pi_\gamma(B) : B \in \mathcal{B}\}$ of subsets of X_γ has the finite intersection property.

Now X_γ is compact which implies that $\bigcap_{B \in \mathcal{B}} \overline{\pi_\gamma(B)} \neq \emptyset$. So there exists $x_\gamma \in X_\gamma$ such that $x_\gamma \in \bigcap_{B \in \mathcal{B}} \overline{\pi_\gamma(B)}$. This means that for each open neighbourhood U_γ of x_γ and all $B \in \mathcal{B}$, we have $U_\gamma \cap \pi_\gamma(B) \neq \emptyset$, and therefore $\pi_\gamma^{-1}(U_\gamma) \cap B \neq \emptyset$. Therefore, by our earlier observation, $\pi_\gamma^{-1}(U_\gamma) \in \mathcal{B}$.

Let $x = (x_\gamma)_{\gamma \in \Gamma}$ (from now on, x is defined with specifically the x_γ above) and let U be an open neighbourhood of x. By definition of the product topology on X, there exist $\gamma_1, \gamma_2, ..., \gamma_n \in \Gamma$ and open neighbourhoods of x_{γ_i} which we denote U_{γ_i} such that $\bigcap_{i=1}^n \pi_{\gamma_i}^{-1}(U_{\gamma_i}) \subseteq U$.

Each $\pi_{\gamma_i}^{-1}(U_{\gamma_i}) \in \mathcal{B}$ so (since we established that $\mathcal{B}$ is closed under finite intersections) $\bigcap_{i=1}^n \pi_{\gamma_i}^{-1}(U_{\gamma_i}) \in \mathcal{B}$ so for all $B \in \mathcal{B}$ we have $\bigcap_{i=1}^n \pi_{\gamma_i}^{-1}(U_{\gamma_i}) \cap B \neq \emptyset$.

Therefore, for all $B \in \mathcal{B}$ we have $U \cap B \neq \emptyset$. U was arbitrary which means every open neighbourhood of x intersects every $B \in \mathcal{B}$ non-trivially, which means $x \in \overline{B}$ for every B as otherwise x would be in an open set not containing B. This means that $\bigcap_{B \in \mathcal{B}} \overline{B}$ is non-empty as required.

□

Remark. We used the axiom of choice twice: Once by invoking Tukey's lemma, and again by picking an x_γ for each γ. Although we won't prove this here, Tychonoff's theorem turns out to be equivalent to the axiom of choice.

Theorem. (Banach-Alaoglu Theorem) Let X be a normed space, then consider the closed unit ball in X^* given by

$$B\left(X^*\right) = \left\{f \in X^* : \|f\| \leq 1\right\}$$

is compact in the weak-star topology.

Proof. Let $\mathbb{K}$ be the underlying field.

For $x \in X$, let $D_x = \{\xi \in \mathbb{K} : |\xi| \leq \|x\|\}$ with the subspace topology. Let $D = \prod_{x \in X} D_x$ with the product topology. Each D_x is compact so by Tychonoff's Theorem, D is compact with respect to the product topology. Write $\xi \in D$ as $(\xi_x)_{x \in X}$.

Define $\phi : B\left(X^*\right) \to D$ with $f \mapsto (f(x))_{x \in X}$. Then for each x, $\|f\| \|x\| \leq x$ so $f(x) \in D_x$ so it actually maps to D.

Note that ϕ is injective because if $\phi(f) = \phi(g)$ then $f(x) = g(x)$ for all x so $f = g$.

Note that ϕ is continuous with respect to the weak star topology because if $U \subseteq D$ is an open neighbourhood of $\phi(f_0) = (f_0(x))_{x \in X}$ then there exist $x_1, x_2, ..., x_n \in X$ and $\varepsilon_1, \varepsilon_2, ..., \varepsilon_n$ such that $\{\xi \in D : \forall i \, |\xi_{x_i} - f_0(x_i)| < \varepsilon_i\} \subseteq U$ by definition of the product topology.

Then

$$\phi^{-1}\left(\{\xi \in D : \forall i \, |\xi_{x_i} - f_0(x_i)| < \varepsilon_i\}\right) = \{f \in B\left(X^*\right) : \forall i \, |f(x_i) - f_0(x_i)| < \varepsilon_i\} \in T_w^*$$

This means that ϕ is indeed continuous.

The proof for ϕ^{-1} is similar.

This means ϕ, although not generally surjective, is a Homeomorphism onto its image.

The result follows if we can prove $\phi\left(B\left(X^*\right)\right)$ is compact, and it is enough to show that it is closed, since D is compact and Hausdorff.

The idea is that we want to show that $\phi\left(B\left(X^*\right)\right) = \overline{\phi\left(B\left(X^*\right)\right)}$.

Suppose $\xi = (\xi_x)_{x \in X} \in \overline{\phi\left(B\left(X^*\right)\right)}$. Define $f : X \to \mathbb{K}, x \mapsto \xi_x$. Then we want to show that $f \in X^*$.

Since $\xi \in \overline{\phi\left(B\left(X^*\right)\right)} \subseteq D$, there exists a sequence $f_n \in B\left(X^*\right)$ such that $\phi(f_n) \to \xi$ i.e. $f_n \to f$ as $n \to \infty$ for all $x \in X$ fixed (since this is equivalent to convergence in product topology). Let $x, y \in X$ and $a, b \in \mathbb{K}$. Then

$$|f(ax + by) - af(x) - bf(y)|$$

$$\leq |f(ax+by) - f_n(ax+by)| + |f_n(ax+by) - af_n(x) - bf_n(y)| + |af_n(x) - af(x)| + |bf_n(y) - bf(y)|$$

Since each f_n is linear and $f_n \to f$ this $\to 0$ as $n \to \infty$ so f is linear.

Since $|f(x)| = |\xi_x| \leq \|x\|$ for all $x \in X$ (since we are in D), f is continuous and $\|f\| \leq 1$ so $f \in B\left(X^*\right)$. By definition of f, we have $\xi = \phi(f) \in \phi\left(B\left(X^*\right)\right)$ which is thus equal to $\overline{\phi\left(B\left(X^*\right)\right)}$.

□

Theorem. (Goldstine's Theorem) Let X be a normed space and let $B(X) = \{x \in X : \|x\| \leq 1\}$ and let $\phi_X : X \to X^{**}$ be defined by $\phi_X(x)(f) = f(x)$. Then $\phi_X(B(X))$ is w^*-dense in $B\left(X^{**}\right)$.

Proof. Let $\mathbb{K}$ be the underlying field.

Let U be w^*-open in $B\left(X^{**}\right)$ and let $\theta \in U$. Then there exist $f_1, f_2, ..., f_n \in X^*$ such that

$$\{\psi \in X^{**} : \forall i : |\theta(f_i) - \psi(f_i)| < 1\}$$

To show that $\phi_X(B(X))$ is dense we want to show that U contains ψ of the form $\phi_X(x)$ for some $x \in B(X)$. Thus it is enough to show that for every $f_1, f_2, ..., f_n$ there exists $x \in B(X)$ such that

$$\sum_{i=1}^{n} |\theta(f_i) - \phi_X(x)(f_i)| = \sum_{i=1}^{n} |\theta(f_i) - f_i(x)| < 1$$

Assume the f_i are linearly independent, because if the statement above is true for every linearly independent set then it is easy to see that it is true in general, as if f_j is linearly independent of the others you can just remove it and rescale the rest of the terms appropriately.

Define $J : X \to \mathbb{K}^n$ by $x \mapsto (f_1(x), f_2(x), ..., f_n(x))$. Then J is linear, continuous (since the f_i's are in X^*) and (by linear independence which prevents zero functions) surjective, but not generally injective. Define an equivalence relation on X by $x \sim y$ if and only if $Jx = Jy$ and let $[x]$ be the equivalence class of x. The set of equivalence classes is exactly the quotient $\hat{X} = X/\ker(J)$. Also define $\|[x]\| = \inf_{y \sim x} \|y\|$.

Now one may define $\hat{J} : \hat{X} \to \mathbb{K}^n$ by $\hat{J}([x]) = J(x)$. Then $\hat{J}$ is a linear bijection so there exists a unique $[z] \in \hat{X}$ such that $\hat{J}([z]) = (\theta(f_1), \theta(f_2), ..., \theta(f_n))$. By definition of $\hat{J}$ this equals $J(z) = (f_1(z), f_2(z), ..., f_n(z))$.

By a corollary of Hahn-Banach there exists $\hat{f} \in \hat{X}^*$ such that $\left\|\hat{f}\right\| = 1$ and $\hat{f}([z]) = \|[z]\|$.

Let $b = \hat{f} \circ \hat{J}^{-1} \in (\mathbb{K}^n)^*$, say $b = (b_1, b_2, ..., b_n)$. Then

$$\theta(b \circ J) = \theta\left(\sum b_i f_i\right) = \sum b_i \theta(f_i) = b \circ \hat{J}([z]) = \hat{f}([z]) = \|[z]\|$$

Also we have

$$|b \circ J(x)| = \left|b \circ \hat{J}([x])\right| = \left|\hat{f}([x])\right| \leq \left\|\hat{f}\right\| \|[x]\| = \|[x]\| \leq \|x\|$$

This implies that $\|b \circ J\| \leq 1$ so

$$\|[z]\| \leq \|\theta\| \|b \circ J\| \leq \|\theta\| \leq 1$$

Now fix $\varepsilon > 0$. Since $\|[z]\| = \inf_{x \sim z} \|x\|$ there exists $x \in X$ such that $x \sim z$ and $\|x\| < 1 + \varepsilon$. Then $\sum |\theta(f_i) - f_i(x)| = 0$ (since $x \sim z$ so each term in the sum is 0) and $\left\|\frac{x}{1+\varepsilon}\right\| < 1$ and consider $\sum \left|\theta(f_i) - f_i\left(\frac{x}{1+\varepsilon}\right)\right|$. Then we want to show that there is an ε such that this is less than 1. We have

$$\sum \left|\theta(f_i) - f_i\left(\frac{x}{1+\varepsilon}\right)\right| = \sum \left|\theta(f_i) - \frac{1}{1+\varepsilon} f_i(x)\right| = \sum \left|\frac{\varepsilon}{1+\varepsilon} f_i(x)\right| < \varepsilon \sum |\theta(f_i)|$$

Since $\sum |\theta(f_i)|$ is fixed in this part of the proof, there is a choice of ε that makes this work.

□

Lemma. Let X be a normed space. Then the Weak-Star topology on X^{**} is Hausdorff.

Proof. Note that X^* separates points in X by a corollary of Hahn-Banach. If two elements $F_1, F_2 \in X^{**}$ are distinct then there must be some point $f \in X^*$ where they disagree meaning $F_1(f) \neq F_2(f)$. Therefore in the underlying field $\mathbb{K}$ there are open sets U_1, U_2 disjoint such that $F_1(f) \in U_1$ and $F_2(f) \in U_2$. Then note that

$$\phi_{X^*}(f)^{-1}(U_1) = \{F \in X^{**} : \phi_{X^*}(f)(F) \in U_1\}$$

$$\phi_{X^*}(f)^{-1}(U_2) = \{F \in X^{**} : \phi_{X^*}(f)(F) \in U_2\}$$

These are both open since the weak star topology is designed to make all the functions $\phi_{X^*}(f)$ continuous, so they are genuine disjoint open neighbourhoods of F_1 and F_2.

□

Lemma. Let X be a normed space. Then the map $\phi_X : (X, T_w) \to (X^{**}, T_w^*)$ is continuous and if X is reflexive then the map $\phi_X^{-1} : (X^{**}, T_w^*) \to (X, T_w)$ is continuous.

Proof. To check that a function is continuous, it is enough to check that pre-images of open sets in the subbasis are open.

An open set in the subbasis of (X^{**}, T_w^*) looks like

$$V_{f,U} = \{F \in X^{**} : F(f) \in U\}$$

where $f \in X^*$ and U is open in the underlying field $\mathbb{K}$. The pre-image of this subbasic open set is

$$\begin{aligned} \phi_X^{-1}(V_{f,U}) &= \{x \in X : \phi_X(x) \in V_{f,u}\} = \{x \in X : \phi_X(x)(f) \in U\} \\ &= \{x \in X : f(x) \in U\} = f^{-1}(U) \end{aligned}$$

Since f is continuous in the weak topology T_w on X by definition, it follows that $f^{-1}(U)$ is open in (X, T_w), so ϕ_X is continuous.

Now let X be reflexive. For ϕ_X^{-1}, note that reflexivity implies ϕ_X is a bijection, so ϕ_X^{-1} is also a bijection.

A subbasic open set in the weak topology on X is defined by

$$W_{f,U} = \{x \in X : f(x) \in U\} = f^{-1}(U)$$

where $f \in X^*$ and U is open in $\mathbb{K}$. We need to show that $\left(\phi_X^{-1}\right)^{-1}(W_{f,U})$ is open in (X^{**}, T_w^*). We have

$$\left(\phi_X^{-1}\right)^{-1}(W_{f,U}) = \phi_X(W_{f,U})$$

since ϕ_X is a bijection. Therefore, this is

$$= \{\phi_X(x) \in X^{**} : f(x) \in U\}$$

Because X is reflexive, every $F \in X$ is equal to $\phi_X(x)$ for some $x \in X$. Therefore write $f(x) = \phi_X(x)(f) = F(f)$. So the set that we want to prove is open is

$$\{F \in X^{**} : F(f) \in U\}$$

And wouldn't you look at that, that's exactly $V_{f,U}$.

□

Corollary. Let X be a Banach space, then X is reflexive if and only if $B(X)$ is w-compact.

Proof. Note that we proved earlier that if X is reflexive then X^* is reflexive. This implies that on X^{**}, $T_w = T_w^*$. Therefore by Banach-Alaoglu, $B(X^{**})$ is w-compact and $B(X) = \phi_X^{-1}(B(X^{**}))$ is also w-compact, since ϕ_X is a homeomorphism by the previous lemma.

Conversely, suppose $B(X)$ is w-compact. Then $\phi_X(B(X))$ is w^*-compact and thus w^*-closed in $B(X^{**})$ as X^{**} is a Hausdorff space. By Goldstine's theorem, $\phi_X(B(X))$ is w^*-dense in $B(X^{**})$. Therefore, since $\phi_X(B(X))$ is closed, this implies that it equals $B(X^{**})$. By linearity it follows that $\phi_X(X) = X^{**}$ so X is reflexive.

□

Definition. A sequence (x_n) in X is said to be weakly Cauchy if for every w-open neighbourhood V of 0 we have $x_n - x_m \in V$ for all m, n large enough.

Theorem. Every w-Cauchy sequence in a reflexive Banach space is w-convergent.

Proof. Let (x_n) be w-Cauchy in X and let $f \in X^*$. Then $f(x_n)$ is Cauchy in $\mathbb{K}$ and thus $f(x_n)$ converges in $\mathbb{K}$. Therefore $f(x_n)$ is bounded in $\mathbb{K}$. In other words, for all f there exists a c_f such that $|\phi_X(x_n)| \leq c_f$ for all n. Thus, by the Banach-Steinhaus uniform boundedness principle there exists $c > 0$ such that $\|x_n\| = \|\phi_X(x_n)\| \leq c$ for all n, since ϕ_X is an isometry.

Define $T : X^* \to \mathbb{K}$ by $f \mapsto \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} \phi_X(x_n)(f)$. Then T is linear and bounded, since $|T(f)| \leq c\|f\|$, so $T \in X^{**}$. Since X is reflexive, $T = \phi_X(x)$ for some $x \in X$ and therefore the limit of $f(x_n)$ is actually in the image of f, i.e. it is $Tf = \phi_X(x) f = f(x)$, so by an earlier criterion we proved we have $x_n \rightharpoonup x$ for this x.

□

Theorem. Let X be a reflexive Banach space. Then $B(X)$ is sequentially compact.

Proof. Let (x_n) be a sequence in $B(x)$. Let $X_1 = \overline{\langle (x_n) \rangle}$. Then X_1 is separable (consider, for example, the set of finite rational linear combinations of the x_i's which is countable) and reflexive. This implies that X_1^{**} is separable so X_1^* is separable. Let (f_n) be a sense sequence in X_1^*.

Now $(f_1(x_n))$ is a bounded sequence in $\mathbb{K}$ so there exists a subsequence $x_{1,j}$ of $x_n$such that $f_1(x_{1,j})$ converges.

Now $(f_2(x_{1,n}))$ is a bounded sequence in $\mathbb{K}$ so there exists a subsequence $x_{2,j}$ of $x_{1,n}$ such that $f_2(x_{2,j})$ converges.

Continuing in this manner, we can find subsequences $x_{j,k}$ for $j, k \in \mathbb{N}$. Let $v_k = x_{k,k}$. It follows that $f_n(v_k)$ converges for all n.

Now we want to show that for arbitrary $f \in X_1^*$, the sequence $(f(v_k))_{k=1}^{\infty}$ is a Cauchy sequence in $\mathbb{K}$. Once we have this, it will follow that v_k is w-Cauchy, since for each basic open neighbourhood of 0 and m, n large enough a finite collection of $f_i \in X_1^*$ will have $f_i(v_m) - f_i(v_n)$ be in the corresponding open neighbourhoods. It will then follow from the previous theorem that v_k is w-convergent which would complete the proof.

Fix $\varepsilon > 0$. By density (in the norm topology of X^*), there is some N such that $\|f - f_N\| < \frac{\varepsilon}{3}$. Then we have, since $\|v_k\|$ is bounded by 1 and $f_N \to f$ that if m, n arge large enough then

$$|f(v_m) - f(v_n)| \leq |f(v_m) - f_N(v_m)| + |f_N(v_m) - f_N(v_n)| + |f_N(v_n) - f(v_n)| < \varepsilon$$

Thus f is Cauchy as required.

$\square$

Corollary. Let X be a reflexive Banach space. Then every bounded sequence in X has a convergent subsequence.

6 Hilbert spaces

6.1 Basic properties

Recall that a Hilbert space is a Banach space equipped with a real or complex/hermitian inner product and an associated norm.

Definition. Given a metric space X, define its completion $\tilde{X}$ by the set of equivalence classes of Cauchy sequences where $(x_n) \sim (y_n)$ if and only if $x_n - y_n \to 0$. Think of making $\mathbb{R}$ out of $\mathbb{Q}$.

The completion is unique up to isomorphism for the same proof as the similar fact earlier, removing all parts of that proof that talked about linearity.

Proposition. If V is an inner product space one can assign a unique inner product to its completion that agrees with the inner product on V.

Proof. We will define $\langle x, y \rangle := \lim_{n\to\infty} \langle x_n, y_n \rangle$ where these are Cauchy sequences converging to x, y respectively. We will prove that (1) the limit exists, (2) that this is independent of the choice of sequence, (3) that this is actually an inner product and (4) that it is unique. Note that by considering constant sequence it agrees with the inner product on V.

Proof of (1):

It suffices to prove that $\langle x_n, y_n \rangle$ is Cauchy. We have, by the triangle and Cauchy-Schwarz inequalities, that

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x_m, y_m \rangle| &\leq |\langle x_n - x_m, y_n \rangle| + |\langle x_m, y_n - y_m \rangle| \\ &\leq \|x_n - x_m\| \|y_n\| + \|y_n - y_m\| \|x_m\| \end{aligned}$$

Since y_n, x_m are both Cauchy and thus bounded and $\|y_n - y_m\|$ and $\|x_n - x_m\|$ tends to 0, the whole thing tends to 0.

Proof of (2):

If we pick different Cauchy sequences x'_n, y'_n that converge to the same limit, then by the same bounding argument as above, $|\langle x_n, y_n \rangle - \langle x'_n, y'_n \rangle|$ tends to 0 so the 2 sequences of inner products approach the same limit.

Proof of (3): Because limits preserve addition and multiplication, bilinearity or sesqilinearity and conjugate symmetry carry over. Positive definiteness follows because limits preserve non-strict inequalities. And $\|x\| = 0 \iff x = 0$ in $\tilde{V}$.

Proof of (4): Recall from Linear Algebra that the polarization identity (both the real and complex versions) means the norm determines the inner product. From the fact that the norm is uniquely determined by the completion of a normed space by a theorem from earlier, the result follows.

□

Definition. A Euclidean space is a vector space with a hermitian inner product

Theorem. Let $\|\cdot\|$ be an inner product norm, let x, y be arbitrary elements of a euclidean space, and let $x_1, x_2, ..., x_n$ be mutually orthogonal (i.e. $\langle x_i, x_j \rangle = 0$) elements of said euclidean space. Then we have the following:

1. (Parallelogram identity) $\|x-y\|^2 + \|x+y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right)$

2. (Pythagoras theorem) $\|x_1 + x_2 + ... + x_n\|^2 = \|x_1\|^2 + \|x_2\|^2 + ... + \|x_n\|^2$

Proof. Easy exercise. (Hint: Use $\|x\| = \langle x, x \rangle$)

□

Definition. Let X be a Euclidean space and $S \subseteq X$ be any subset, then define its orthogonal complement as

$$S^{\perp} = \{y \in X : \forall z \in S, \langle y, z \rangle = 0\}$$

Proposition. Orthogonal complements are always closed.

Proof. Since $S^{\perp} = \bigcap_{x \in S} z^{\perp}$ it suffices to prove that each $x^{\perp}$ is closed.

Let x_n be such that $\langle x_n, x \rangle = 0$ and suppose that $x_n \to y$, then $|\langle y - x_n, x \rangle| \le \|y - x_n\| \|x\| \to 0$ by Cauchy-Schwarz. Thus, $x^{\perp}$ contains all its limit points because the sequence was arbitrary.

□

Observe that in general, $A \subseteq B \implies B^{\perp} \subseteq A^{\perp}$. For any subset S, define $V = \overline{\text{lin}}(S)$ to be the closure of the set of finite linear combinations of S. Then $V^{\perp} \subseteq S^{\perp}$. We then have equality since something perpendicular to all of S is perpendicular to all of V.

Note that clearly, $S \subseteq \left(S^{\perp}\right)^{\perp}$ because everything in S is perpendicular to everything in $S^{\perp}$. Becuase $S^{\perp} = V^{\perp}$ their orthogonal complements are equal. Therefore we have $\left(S^{\perp}\right)^{\perp} = \left(V^{\perp}\right)^{\perp}$. By a theorem that we are shortly going to prove, if we are in a Hilbert space then we will have $\left(V^{\perp}\right)^{\perp} \subseteq V$, so $\left(S^{\perp}\right)^{\perp} = \text{lin}(S)$.

Theorem. Let X be a Euclidean space and let $Y \subseteq X$ be a complete subspace. Then every $x \in X$ can be written uniquely as $y + y^*$ where $y \in Y$ and $y^* \in Y^{\perp}$.

Furthermore, the projections P_Y and $P_{Y^{\perp}}$ are bounded, self-adjoint linear operators with norm ≤ 1.

Proof. **Step 1.** Existence.

Let $d = \inf_{y \in Y} \|x-y\|$. Take a sequence $(y_n)_{n=1}^{\infty} \subseteq Y$ with $\|x-y_n\|^2 \leq d^2 + \frac{1}{n}$.

We will prove that the sequence as defined above is Cauchy, Recall the parallelogram law which says $\|x-y\|^2 + \|x+y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right)$. Apply this with $x - y_n$ and $x - y_m$ to get

$$\|y_n - y_m\|^2 = 2\|x-y_n\|^2 + 2\|x-y_m\|^2 - \|2x - y_n - y_m\|^2$$

$$\leq 2\left(d^2 + \frac{1}{n}\right) + 2\left(d^2 + \frac{1}{m}\right) - 4d^2$$

The last term is because $\|2x - y_n - y_m\|^2$ is 4 times as big as $\left\|x - \frac{y_n + y_m}{2}\right\|^2 = d^2$.

This is $2\left(\frac{1}{n} + \frac{1}{m}\right)$ so the sequence is Cauchy.

Since Y is complete it follows that the limit, y, is in Y and has $\|x-y\| = d$.

We need to prove that $y^* := (x-y) \perp Y$. Suppose on the contrary that there exists $z \in Y$ with $\langle y^*, z\rangle \neq 0$. Multiplying z by some $e^{i\theta}$ if necessary we may assume that $\langle y^*, z\rangle$ is real and is > 0.

Let $y' = y + \varepsilon z$ for $\varepsilon > 0$. Then $y' \in Y$ and we have

$$\|x - y'\|^2 = \langle x - (y + \varepsilon z), x - (y + \varepsilon z)\rangle$$

$$= \langle y^* - \varepsilon z, y^* - \varepsilon z\rangle = \|y^*\|^2 - 2\varepsilon\langle y^*, z\rangle + \varepsilon^2\|z\|^2$$

Taking ε sufficiently small (since the middle term is non-zero so we are not at a local minimum of the quadratic in ε) this contradicts minimality of y in terms of distance in Y to x.

Step 2. Uniqueness.

Suppose $x = y_1 + y_1^* = y_2 + y_2^*$. Then $0 = y + y^*$ for some non-zero $y = y_1 - y_2$ and $y^* = y_1^* - y_2^*$. This makes y orthogonal to minus itself which is a contradiction.

Step 3. The part about the projections.

Clearly they are linear.

Boundedness follows from Pythagoras: $\|x\|^2 = \|P_Y x\|^2 + \|P_{Y^\perp} x\|^2$, so $\|P_Y x\| \leq \|x\|$ and $\|P_{Y^\perp} x\| \leq \|x\|$.

Self adjointness is because

$$\langle x, P_Y x'\rangle = \langle P_Y x, P_Y x'\rangle = \langle P_Y x, x'\rangle$$

For $P_Y^\perp$ it's similar.

$\square$

Note that if X is a Hilbert space it suffices for Y to be a closed subspace.

Corollary. Suppose X is a Hilbert space and Y is a closed subspace. Then $\left(Y^\perp\right)^\perp = Y$.

Proof. Let $w \in \left(Y^\perp\right)^\perp$, then we want to show that $w \in Y$, as we know from earlier that $Y \subseteq \left(Y^\perp\right)^\perp$. Note that $w = y + y^*$ with $y \in Y$ and $y^* \perp Y$ by the theorem above, and we have $\langle w, y^*\rangle = 0$ because $w \in \left(Y^\perp\right)^\perp$. On the other hand, we have $\langle w, y^*\rangle = \langle y, y^*\rangle + \langle y^*, y^*\rangle$ so $\langle y^*, y^*\rangle = 0$. This establishes that $y^* = 0$ so $w \in Y$ as required.

$\square$

Theorem. (Riesz representation theorem) Let X be a hilbert space and let $\phi : X \to \mathbb{C}$ be a bounded linear functional. Then there exists $x_0 \in X$ such that $\phi(x) = \langle x, x_0\rangle$.

Furthermore, any functional of this type is a bounded linear functional and identification $f : x_0 \to \phi$ gives a bijection $X \to X^*$. This is an anti-isometry in the sense that $f(\lambda x + \mu y) = \bar{\lambda} f(x) + \bar{\mu} f(y)$.

Furthermore, $\|\phi\|_{X^*} = \|x_0\|_X$.

This induces an inner product on X^* by $\langle \phi, \psi \rangle_{X^*} = \langle x_\phi, x_\psi \rangle_X$, with the order changed because it is an anti-isometry.

Proof. Let $Y = \ker(\phi)$. This is a closed subspace of X so $X = Y \oplus Y^\perp$. Note that $\dim\left(Y^\perp\right) \leq 1$ since $\phi|_{Y^\perp}$ has trivial kernel. If $\phi = 0$ then the whole theorem follows so assume $\dim\left(Y^\perp\right) = 1$.

Let $Y^\perp = \langle z \rangle$ for some $z \in X$ and let $x = y + \lambda z$ be an arbitrary element of X. Then $\phi(x) = \lambda \phi(z)$ and $\langle x, z \rangle = \langle \lambda z, z \rangle = \lambda \|z\|^2$

Note that unfortunately, the convention of whether an inner product is linear in the first or second argument is not consistent between texts and therefore will not be consistent in my notes as I pull from many different sources. In these notes, it is conjugate linear in the second argument and linear in the first, but the convention used in my notes depends on the specific sublevel.

Now define $x_0 = \frac{z}{\|z\|^2} \overline{\phi(z)}$ so we get

$$\langle x, x_0 \rangle = \frac{\phi(z)}{\|z\|^2} \langle x, z \rangle = \lambda \phi(z) = \phi(x)$$

Anything of this type is a bounded linear functional by Cauchy-Schwarz and f is an anti-isometry by conjugate linearity in the second argument. As for the norm equality,

$$\|\phi\| = \sup_{\|x\|=1} |\langle x, x_0 \rangle| \leq \sup_{\|x\|=1} \|x\| \|x_0\| = \|x_0\|$$

On the other hand,

$$|\phi(x_0)| = \langle x_0, x_0 \rangle = \|x_0\|^2$$

Thus $\|\phi\| \geq \|x_0\|$ as well.

$\square$

Corollary. Every real Hilbert space is isometrically isomorphic to its own dual and every Hilbert space is isometrically isomorphic to its second dual.

Recall that if $T : X \to Y$ is a map between two normed spaces then we can define the adjoint map $T^* : Y^* \to X^*$ by $(T^* f)(x) = f(Tx)$. Suppose now that $X = Y$ and that they are both Hilbert spaces and that X, X^* have been identified using the Riesz representation theorem.

Then $\langle Tx, x_0 \rangle = (T^* x_0)(x) = \langle x, T^* x_0 \rangle$. That is, we always have $\langle Tx, y \rangle = \langle x, T^* y \rangle$. Note that this holds for hermitian conjugates in finite dimensional complex vector spaces, so the adjoint of a linear map is its Hermitian conjugate.

Note that $\|T^*\| \leq \|T\|$ in general normed spaces as we showed before we used Hahn-Banach to show that equality occurs. But since $T = T^{**}$ in a Hilbert space, this gives a proof of equality in Hilbert spaces that does not require Hahn-Banach.

Note that the completion of the space of continuous functions on $[a, b]$ with the inner product $\langle f, g \rangle = \int f \bar{g}$ and induced norm is $L^2[a, b]$. The reason is because of a lemma I proved in my probability and measure notes that says that continuous functions with compact support are dense in L^p. However, given the tools we have this only generalizes to within $\mathbb{R}^n$. For general metric spaces we define $L^2[X]$ as the completion of the space of continuous functions on X

6.2 Von Neumann's L^2 Ergodic theorem

This will be similar but slightly different to the version from the probability and measure course.

Theorem. Let H be a Hilbert space and let $T : H \to H$ be linear and have norm at most 1. Let Y be the closed subspace of H consisting of vectors with $Ty = y$ and let $\pi : H \to Y$ be the orthogonal projection onto Y. Write $S_N(x) = \frac{1}{N}\left(x + Tx + ... + T^{N-1}x\right)$. Then $S_N(x) \to \pi(x)$ as $N \to \infty$.

Proof. Note that $\|T^*\| \leq 1$ as well. Suppose that x is such that $Tx = x$, then

$$\|x - T^*x\|^2 = \langle x - Tx, x\rangle + \langle x, x - Tx\rangle - \|x\|^2 + \|T^*x\|^2 = \|T^*x\|^2 - \|x\|^2 \leq 0$$

Thus $T^*x = x$.

Suppose $f \in Y$ and define $\partial g := g - Tg$. Let M be the closed subspace spanned by all ∂g for $g \in H$.

Then for all $g \in H$ we have

$$\langle f - \partial g\rangle = \langle f, g\rangle - \langle f, Tg\rangle = \langle f, g\rangle - \langle T^*f, g\rangle = 0$$

since $f \in Y$ and is thus T-invariant.

Therefore $f \in Y$ implies f is orthogonal to M.

Now supposes that f is orthogonal to M, then $\langle f, \partial f\rangle = 0$, but we also have

$$\|f - Tf\|^2 = \langle f, f - Tf\rangle + \langle f - Tf, f\rangle - \|f\|^2 + \|Tf\|^2 \leq 0$$

Therefore $f = Tf$ so $f \in Y$.

This implies that $(\text{im}(I - T))^\perp = Y$. Note that the image of a bounded linear operator in general Hilbert spaces is not necessarily closed. We have $(\text{im}(I - T))^{\perp\perp} = Y^\perp$ so $Y^\perp = \overline{\text{lin}}(\text{im}(I - T))$, by properties of the double complement from earlier.

Let $x \in H$ and $\varepsilon > 0$. Since $H = Y + Y^\perp$ we can write $x = \pi(x) + \partial g + h$ where $\|h\| < \frac{\varepsilon}{2}$. Then $S_N(\pi(x)) = \pi(x)$, and $\|S_n h\| < \frac{\varepsilon}{2}$. It remains to check that $\|S_n \partial g\| \leq \frac{\varepsilon}{2}$ then we will be done. Indeed,

$$S_N \partial g = \frac{1}{N}\left((g - Tg) + \left(Tg - T^2 g\right) + ... + \left(T^{N-1}g + T^N g\right)\right) = \frac{1}{N}\left(g - T^N g\right)$$

So we have, for N sufficiently large,

$$\|S_N \partial g\| \leq \frac{2}{N}\|g\| < \frac{\varepsilon}{2}$$

as required.

□

6.3 Orthonormal systems and Fourier series

Let X be a Euclidean space. Then a collection of elements $(\phi_i)_{i\in I}$ is said to be orthonormal if for all $i, j \in I$ we have $\langle \phi_i, \phi_j\rangle = \delta_{ij}$.

Let X be a Hilbert space and let (ϕ_i) be an orthonormal system. Then we say it is complete if we can't add another element and still have an orthonormal system.

Proposition. An orthonormal system $(\phi_i)_{i\in I}$ is complete if and only if its closed linear span is X.

Proof. Suppose $(\phi_i)_{i\in I}$ has closed linear span X. Then $\left((\phi_i)_{i\in I}\right)^\perp = X^\perp = \{0\}$.

Conversely, suppose $(\phi_i)_{i\in I}$ is complete and let Y be its closed linear span. Then $Y^\perp = 0$. But $X = Y \oplus Y^\perp$ so $X = Y$.

□

Recall Gram-Schmidt orthonormalisation from Linear Algebra. The statement is that if $x_1, x_2, ...$ is a linearly independent countable set then there exist orthonormal vectors $y_1, y_2, ...$ such that for all k the spans $\langle x_1, x_2, ..., x_k \rangle = \langle y_1, y_2, ..., y_k \rangle$.

Note that the orthogonal projection of X onto $\langle y_1, y_2, ..., y_k \rangle$ is given by

$$\pi(x) = \langle x, y_1 \rangle y_1 + \langle x, y_2 \rangle y_2 + ... + \langle x, y_k \rangle y_k$$

It is easy to check this.

Corollary. , Let X be a Hilbert space with a countable subset with dense linear span. Then X is a separable Hilbert space and X has a complete orthonormal system that is countable (possibly finite).

Proof. Take finite rational linear combinations of the countable subset to get X is separable.

Apply Gram-Schmidt to the countable subset with dense linear span to get the orthonormal system.

□

Example. l_2 is separable and the orthonormal system $e_i = (0, 0, ..., 0, 1, 0, ...)$ is complete.

Example. Define $L^2(\pi)$ to be the completion of L^2-norm of continuous functions on the unit circle, or equivalently, of 2π-periodic functions.

Then the system $(\phi_n) = \frac{1}{\sqrt{2\pi}} e^{inx}$ is orthonormal - we checked this earlier.

Less obviously, (ϕ_n) is complete. In L^∞ this follows from an application of Stone-Weierstrass from earlier. Since uniform convergence implies L^2 convergence (easy check) it actually follows that (ϕ_n) is complete in L^2 as well.

The following theorem implies that $L^2(\pi)$ is isomorphic to l^2.

Theorem. (Riesz-Fischer Theorem) Suppose $(\phi_n)_{n=1}^\infty$ is an orthonormal system in a Hilbert space X and $(a_n)_{n=1}^\infty$ is a sequence of complex numbers, then $\sum_{n=1}^\infty a_n \phi_n \in X$ if and only if $\sum |a_n|^2 < \infty$. Also, we have $\|\sum_{n=1}^\infty a_n \phi_n\| = \sqrt{\sum |a_n|^2}$

Proof. Write $x_N = \sum_{n=1}^N a_n \phi_n$. Then $\|x_N - x_M\|^2$ for $N > M$ is $\sum_{n=M+1}^N |a_n|^2$ by Pythagoras, so indeed x_N converges if and only if $\sum |a_n|^2$ does.

Since norms preserves limits the equality $\|\sum_{n=1}^\infty a_n \phi_n\| = \sqrt{\sum |a_n|^2}$ follows by Pythagoras.

□

Let ϕ_n be any orthonormal system in a Hilbert space X, then if $x = \sum_{n=1}^\infty c_n \phi_n$ then the c_n are called the Fourier coefficients of x with respect to ϕ_n. We also have $c_n = \langle x, \phi_n \rangle$ since limits preserve orthogonality.

Lemma. Let X be a hilbert space and let $(\phi_n)_{n=1}^\infty$ be an orthonormal system. Let V be the closed subspace spanned by ϕ so that if ϕ_n is complete then $V = X$.

Then $\sum_{n=1}^\infty \langle x, \phi_n \rangle \phi_n = \pi_V x$ where π_V is the orthogonal projection onto V.

Proof. First, we check that this infinite sum actually makes sense.

Write $x_m = \sum_{n=1}^m \langle x, \phi_n \rangle \phi_n$, then $x - x_m$ is orthogonal to every $\phi_1, \phi_2, ..., \phi_m$. Therefore $x = x_m + (x - x_m)$ is an orthogonal decomposition so $\|x_m\| \le \|x\|$. But $\|x_m\|^2 = \sum_{n=1}^m |\langle x, \phi_n \rangle|^2$ which implies that $\sum_{n=1}^\infty |\langle x, \phi_n \rangle|^2$ is a convergent sum. Therefore, $\sum_{n=1}^\infty \langle x, \phi_n \rangle \phi_n$ actually exists.

We now know that $x - \sum_{n=1}^m \langle x, \phi_n \rangle \phi_n = \lim_{m \to \infty} (x - x_m)$. This is orthogonal to the first m ϕ's.

Therefore, $x - \sum_{n=1}^{\infty} \langle x, \phi_n \rangle \phi_n$ is orthogonal to every ϕ_n , since orthogonality is preserved by limits.

Therefore, $\sum_{n=1}^{\infty} \langle x, \phi_n \rangle \phi_n$ is the component in V of the orthogonal decomposition of x. In other words, it is indeed $\pi_V x$ by uniqueness of orthogonal decomposition.

□

Corollary. (Bessel's Inequality) $\sum_{n=1}^{\infty} |\langle x, \phi_n \rangle|^2 = \|\pi_V x\|^2 \leq \|x\|^2$

Lemma. If $x, y \in X$ then

$$\sum_{n=1}^{\infty} \langle x, \phi_n \rangle \overline{\langle y, \phi_n \rangle} = \langle \pi_V x, \pi_V y \rangle = \langle \pi_V x, y \rangle = \langle x, \pi_V y \rangle$$

Proof. Note that $\langle \pi_V x, \pi_V y \rangle = \lim_{m \to \infty} \langle x_m, y_m \rangle$ where x_m is defined as above and y_m is defined similarly.

Therefore,

$$\langle \pi_V x, \pi_V y \rangle = \lim_{m \to \infty} \sum_{n=1}^{m} \langle \langle x, \phi_n \rangle \phi_n, \langle y, \phi_n \rangle \phi_n \rangle = \lim_{m \to \infty} \sum_{n=1}^{m} \langle x, \phi_n \rangle \overline{\langle y, \phi_n \rangle} = \sum_{n=1}^{\infty} \langle x, \phi_n \rangle \overline{\langle y, \phi_n \rangle}$$

Since projection is self-adjoint as we proved earlier, the other inequalities follow.

□

Corollary. (Parseval's identities) If $(\phi_n)_{n=1}^{\infty}$ is complete then $\pi_V = I$ so we get $\sum_{n=1}^{\infty} \langle x, \phi_n \rangle \overline{\langle y, \phi_n \rangle} = \langle x, y \rangle$ and in particular $\sum_{n=1}^{\infty} |\langle x, \phi_n \rangle|^2 = \|x\|^2$

Corollary. If X has a complete orthonormal system (ϕ_n) then X is isometrically isomorphic to l_2 with the isometry $X \to l_2$ given by the map $x \mapsto (\langle x, \phi_n \rangle)_{n=1}^{\infty}$.

Theorem. Fourier series converge in L^2 for continuous functions. Specifically, let f be a 2π-periodic measurable function and suppose that $\int_0^{2\pi} |f|^2$ is finite and let $S_N f = \sum_{n=-N}^{N} \hat{f}(n) e^{inx}$, then $\|S_N f - f\|_2 \to 0$.

Proof. For simplicity, let the inner product be $\langle f, g \rangle = \frac{1}{2\pi} \int f\bar{g}$ instead of $\int f\bar{g}$ so that e^{inx} becomes orthonormal.

Work in the complete Hilbert space $L^2(\pi)$ from earlier and note that consequences of Stone-Weierstrass discussed earlier imply that e^{inx} is indeed a complete orthonormal system.

Let $f(n) = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx = \langle f, \phi_n \rangle$

Thus $\|S_N f - f\|_2 = \sum_{|n| > N} |\langle f - \phi_n \rangle|^2 \to 0$ as $N \to \infty$.

□

This does not contradict the earlier example since convergence in L^2 does not imply pointwise convergence.

Note that often we may want Fourier series to converge pointwise. Although this is not true in general as we showed earlier, it is true at all points of differentiability of the function in question.

To prove this, I will state the Riemann-Lebesgue lemma, something which we will prove in Analysis of functions and assume here.

Theorem. (Riemann-Lebesgue lemma) Let $f : \mathbb{R} \to \mathbb{C}$ be in $L^1(\mathbb{R})$. Then as $|\lambda| \to \infty$ we have $\int_{\mathbb{R}} f(t) e^{i\lambda t} dt \to 0$.

By considering real and imaginary parts and restricting to $[a,b]$ we get

$$\lim_{|\lambda|\to\infty}\int_a^b f(t)\cos(\lambda t)\,dt=0 \text{ and } \lim_{|\lambda|\to\infty}\int_a^b f(t)\sin(\lambda t)\,dt=0$$

Theorem. (Dini's criterion) Let $f:\mathbb{R}\to\mathbb{C}$ be 2π-periodic and Lebesgue integrable on $[0,2\pi]$. For a fixed $x\in\mathbb{R}$ and candidate function S, if there exists a $\delta>0$ such that

$$\int_0^\delta \frac{|f(x+t)+f(x-t)-2S|}{t}dt<\infty$$

then the Fourier series of f evaluated at x converges to S.

In particular, this implies that the Fourier series converges to f at all points where f is differentiable, and that if f has a jump continuity but is continuous near that discontinuity, then as long as f is sufficiently well behaved near the discontinuity (i.e. the criterion holds with $2S=\lim_{t\to0^+}f(x+t)+f(x-t)$, which works if f has finite slope at the discontinuity point[2]), its Fourier series at the discontinuity will converge to the average of the limit point before and after the discontinuity.

Now I will prove the criterion.

Proof. Note that

$$S_N(x)=\sum_{n=-N}^{N}\hat{f}(-n)e^{inx}=\sum_{n=-N}^{N}\frac{1}{2\pi}\int_{-\pi}^{\pi}f(y)e^{-iny}dye^{inx}=\sum_{n=-N}^{N}\frac{1}{2\pi}\int_{-\pi}^{\pi}f(y)e^{-in(x-y)}dy$$

Set $y=x-t$ then we get

$$=\sum_{n=-N}^{N}\frac{1}{2\pi}\int_{x-\pi}^{x+\pi}f(x-t)e^{-int}dt$$

By 2π-periodicity and the fact from earlier that $\sum_{k=-N}^{N}e^{ikx}=D_N(x)=\frac{\sin\left(\left(N+\frac{1}{2}\right)x\right)}{\sin\left(\frac{1}{2}x\right)}$

Thus,

$$S_N(x)=\int_{-\pi}^{\pi}f(x-t)\frac{\sin\left(\left(N+\frac{1}{2}\right)t\right)}{\sin\left(\frac{1}{2}t\right)}dt$$

Now note that $\frac{1}{2\pi}\int_{-\pi}^{\pi}D_N(t)\,dt=1$ so $S=\frac{1}{2\pi}\int_{-\pi}^{\pi}SD_N(t)\,dt$. Therefore

$$S_N(x)-S=\frac{1}{2\pi}\int_{-\pi}^{\pi}[f(x-t)-S]\,D_N(t)\,dt$$

Since $D_N(t)$ is an even function (meaning $D_N(t)=D_N(-t)$) we can "fold over" the above integral so we have

$$S_N(x)-S=\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)-2S]\,D_N(t)\,dt$$

Now let δ be as in the statement of the theorem, i.e. small enough so that integral converges. Define for $0<t\le\pi$

$$g(t)=\frac{f(x+t)+f(x-t)-2S}{2\pi\sin\left(\frac{t}{2}\right)}$$

Our next goal is to prove that $g\in L^1([0,\pi])$. To do this, split the interval.

On $(0,\delta)$, note that since $\delta\le\pi$ there exists a constant C such that $\frac{1}{\sin(t/2)}\le\frac{C}{t}$ for all $t\in(0,\delta)$. Therefore,

$$|g(t)|\le C\frac{|f(x+t)+f(x-t)-2S|}{2\pi t}$$

[2] Yes I know it's not differentiable since its discontinuous and I haven't explained precisely what I mean by finite slope but just use your common sense.

By the hypothesis of the theorem g is Lebesgue integrable on $[0, \delta]$.

On $[\delta, \pi]$, the function $\sin\left(\frac{t}{2}\right)$ is strictly positive and bounded below by $\sin\left(\frac{\delta}{2}\right) > 0$ so $|g(t)| \leq \frac{|f(x+t)+f(x-t)-2S|}{2\pi \sin\left(\frac{\delta}{2}\right)}$, and thus integrable since $f \in L^1$.

Thus, g is integrable on $[0, \pi]$.

Now observe that we can rewrite the error term as

$$S_N(x) - S = \int_0^\pi g(t) \sin\left(\left(N + \frac{1}{2}\right) t\right) dt$$

Since $g \in L^1$, the result follows by the Riemann-Lebesgue lemma. Once we prove that in the analysis of functions course, that will close the gap and we will know that Fourier series converge for probably all cases that the methods course will care about.

□

7 Spectral theory

7.1 Basic properties of the spectrum

The point of this chapter is to generalize the idea of eigenvalues and eigenvectors to linear operators $T : X \to X$ where X is a Banach space.

Definition. If $Tx = \lambda x$ and $x \neq 0$ then x is an eigenvector of eigenvalue λ.

Example. Let $X = l_2$ and define the right shift map by

$$T(x_1, x_2, ...) = (0, x_1, x_2, ...)$$

Then this has no eigenvalues because if $(0, x_1, x_2, ...) = \lambda(x_1, x_2, ...)$ then an easy induction shows that $x_1 = x_2 = ... = 0$

Example. Again let $X = l_2$ and define the left shift map by

$$T(x_1, x_2, ...) = (x_2, x_3, ...)$$

Then every $\lambda \in \mathbb{C}$ with $|\lambda| < 1$ is an eigenvalue with eigenvector $\left(1, \lambda, \lambda^2, \lambda^3, ...\right)$

Note that if $Tx = \lambda x$ then $x \in \ker(T - \lambda I)$ so T is not invertible. However, $T - \lambda I$ does not generally imply λ is an eigenvalue - consider the left shift map above.

Definition. Let X be a Banach space and let $T : X \to X$ be a linear map. (By the way in this chapter from now on if I say T then I implicitly mean this) Then the spectrum of T, denoted $\sigma(T)$ is defined to be

$$\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}$$

If λ is an eigenvalue then we say that λ is in the point spectrum, denoted $\sigma_p(T)$, and $\sigma_p(T) \subseteq \sigma(T)$ with equality if X is finite dimensional (by Linear Algebra work we have done in the past).

Lemma. If $\|T\| < 1$ then $I - T$ is invertible with inverse $I + T + T^2 + T^3 + ...$

Proof. The operator $I + T + T^2 + T^3 + ...$ is well defined - easy to check that the partial sums form a Cauchy sequence by the triangle inequality and properties of geometric series.

Now we have

$$(I - T)\left(I + T + T^2 + T^3 + ...\right) = \lim_{n\to\infty} \left(I - T^{n+1}\right) = I$$

The proof that it is a left inverse is similar.

□

Proposition. $\sigma(T)$ is a compact subset of $\{z \in \mathbb{C} : |z| \leq \|T\|\}$.

Proof. First note that if $\lambda > \|T\|$then $I - \frac{1}{\lambda}T$ is invertible by the lemma above, and thus so is $T - \lambda I$. Therefore, $\sigma(T)$ is a subset of $\{z \in \mathbb{C} : |z| \leq \|T\|\}$. It remains to check that it is compact. To do this, it suffices to check that its complement is open.

Let T be invertible. If S is another operator write $S = T\left(I + T^{-1}(S - T)\right)$. Then we will show that if S is in a certain open ball around T then it is invertible. Specifically, if $\|S - T\| < \left\|T^{-1}\right\|^{-1}$ then $\left\|T^{-1}(S - T)\right\| < \left\|T^{-1}\right\| \|S - T\| \leq 1$, so by the previous lemma it follows that S is invertible.

$\square$

Definition. We say λ is an approximate eigenvalue of T if there exists a sequence $(x_n)_{n=1}^{\infty}$ of unit vectors in X with $(T - \lambda I)x_n \to 0$. Note that this does not imply that we can get an eigenvector by the limit of the sequence, since I never said the sequence converges (see the example below).

Example. Consider the map $T : l_2 \to l_2$ that sends

$$(x_1, x_2, x_3, ...) \mapsto \left(x_1, \frac{1}{2}x_2, \frac{1}{3}x_3, ...\right)$$

Then clearly 0 is not an eigenvalue since no sequence that is non-zero gets sent to 0. However, it is an approximate eigenvalue because if $s_n = (0, 0, ..., 0, 1, 0, ...)$ with a 1 in the n'th position, then $Ts_n \to 0$.

Proposition. Let $T : X \to X$ be a bounded linear operator between Banach spaces and let $\lambda \in \sigma(T)$. Then if $T - \lambda I$ has dense image then it is an approximate eigenvalue.

Proof. Replacing T with $T - \lambda I$ we can assume $\lambda = 0$. If 0 is not an approximate eigenvalue for T then there does not exist a sequence of unit vectors x_n with $Tx_n \to 0$. Thus, T is bounded below, so $\|Tx\| \geq \varepsilon \|x\|$ for some $\varepsilon > 0$.

Thus, T is injective. Assume for a contradiction that T also has dense image and let $Y = \operatorname{im}(T)$. Take $z \in X$ and let $y_1, y_2, ...$ be a sequence of elements in Y with $y_n \to z$. Then T^{-1} restricted to Y is bounded with norm at most $\frac{1}{\varepsilon}$. Therefore, $T^{-1}y_n$ converges to some x since T^{-1} is continuous. But then

$$Tx = \lim_{n\to\infty} T\left(T^{-1}y_n\right) = \lim_{n\to\infty} y_n = z$$

Thus, it follows that T is surjective, and thus T is invertible, so 0 is not in $\sigma(T)$ contradicting the assumption of the theorem.

$\square$

Note that the converse is clear: If $T - \lambda I$ doesn't have dense image then $\lambda \in \sigma(T)$. Also, if λ is an approximate eigenvalue then by the opposite argument to the proof above, it follows that $(T - \lambda I)^{-1}$ cannot be bounded, and we proved earlier that if a bounded linear operator $T - \lambda I$ is bijective then its inverse is bounded, so approximate eigenvalues must be in $\sigma(T)$.

7.2 Compact operators

Let X, Y be Banach spaces and let $T : X \to Y$ be a bounded linear operator.

Definition. T is said to be compact if $\overline{T(\{x \in X : \|x\| \leq 1\})}$ is compact in Y, or equivalently if $T(B_X(1))$ is precompact. Note that $B_X(1)$ is a *closed* ball.

It follows from the open mapping theorem that $T(B_X(1))$ is closed so the "precompact" can be replaced by "compact".

Theorem. (Riesz's Lemma) Let X be a normed space and let $Y \subsetneq X$ be a proper closed subspace and let $\theta \in (0, 1)$. Then there exists a vector $x \in X$ with $\|x\| = 1$ such that $\operatorname{dist}(x, Y) \geq \theta$.

Proof. Since Y is a proper subspace we can choose $u \in X \setminus Y$. Then because Y is closed, $d = \text{dist}\,(u, Y) > 0$. Since d is the infimum and $\theta \in (0,1)$ we know that there exists $y_0 \in Y$ such that $d \leq \|u - y_0\| < \frac{d}{\theta}$. Now let $x = \frac{u-y_0}{\|u-y_0\|}$, then $\|x\| = 1$ and for arbitrary $y \in Y$ we have

$$\|x - y\| = \left\| \frac{u - y_0}{\|u - y_0\|} - y \right\| = \frac{1}{\|u - y_0\|} \|u - y_0 - \|u - y_0\| \, y\|$$

Note that $y' := (y_0 - \|u - y_0\| \, y) \in Y$ so

$$\|x - y\| = \frac{\|u - y'\|}{\|u - y_0\|} > \frac{d}{d/\theta} = \theta$$

Since this holds for all $y \in Y$ it follows that for this x, $\text{dist}\,(x, Y) \geq \theta$.

□

Corollary. $B_X\,(1)$ is compact if and only if X is finite dimensional.

Proof. We know that if X is finite-dimensional then $B_X\,(1)$ is compact.

If X is infinite dimensional then let $x_1 \in X \setminus 0$ and rescale so that $\|x_1\| = 1$. Now let $Y_1 = \langle x_1 \rangle$. Inductively, by the previous lemma, if Y_{k-1} and x_{k-1} are known, pick x_k such that $\|x_k\| = 1$ and $d\,(Y_{k-1}, x_k) \geq \frac{1}{2}$ and set $Y_k = \langle x_1, x_2, ..., x_k \rangle$. Then note that a $\frac{1}{2}$-open ball around each x_k plus the complement of all $\frac{1}{3}$-closed balls around each x_k is an open cover with no finite subcover.

Technical detail: Note that every Y_k is closed due to being a finite dimensional subspace and thus the norm being restricted to Y_k and its natural isomorphism to $\mathbb{K}^k$ sending $a_1x_1 + a_2x_2 + ... + a_kx_k$ to $(a_1, a_2, ..., a_k)$ is equivalent to the standard norm, so the subspace is complete, thus closed.

□

Proposition. Let $X = C\,[0,1]$ and let $K : [0,1]^2 \to \mathbb{C}$ be continuous. Define $T : X \to X$ by $Tf\,(x) = \int_0^1 f\,(y)\,K\,(x,y)\,dy$. Then T is a compact operator on X.

Proof. We need to show that the set $\{Tf : \|f\|_\infty \leq 1\}$ is precompact in X. By the Arzela-Ascoli theorem it suffices to show that it's uniformly bounded and equicontinuous.

For uniform boundedness: Note that

$$\|Tf\|_\infty \leq \|f\|_\infty \sup_x \int_0^1 |K\,(x,y)|\,dy$$

These suprema are finite by compactness. Thus $\{Tf : \|f\|_\infty \leq 1\}$ is uniformly bounded.

For equicontinuity:

$$\begin{aligned} |Tf\,(x_1) - Tf\,(x_2)| &= \left| \int_0^1 f\,(y)\,(K\,(x_1,y) - K\,(x_2,y))\,dy \right| \\ &\leq \|f\|_\infty \left| \int_0^1 (K\,(x_1,y) - K\,(x_2,y))\,dy \right| \end{aligned}$$

By uniform continuity the integral goes to 0 as $x_1 - x_2 \to 0$. Also, $\|f\|_\infty$ has a uniform bound of 1 so we indeed have that for each $\varepsilon > 0$ there is a fixed δ such that $|x_1 - x_2| < \delta$ implies $|Tf\,(x_1) - Tf\,(x_2)| < \varepsilon$.

□

Let $T : l_2 \to l_2$ be a bounded linear operator and define e_n to be the sequence with a 1 in the n'th position and 0 in the other positions and define $\|T\|_{HS} = \sum_{i=1}^\infty \|Te_i\|^2$ if this is finite, in which case we say T is Hilbert-Schmidt.

Proposition. The value $\|T\|_{HS}$ does not depend on the choice of orthonormal basis.

Proof. Let $\hat{e}_i$ be another orthonormal basis. Also let e_i' be another orthonormal basis. Then by Parseval's identity,

$$\|Te_i\|^2 = \sum_{j=1}^{\infty} \left|\langle Te_i, e_j'\rangle\right|^2 = \sum_{j=1}^{\infty} \left|\langle e_i, T^* e_j'\rangle\right|^2$$

Therefore

$$\sum_{i=1}^{\infty} \|Te_i\|^2 = \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} \left|\langle Te_i, e_j'\rangle\right|^2 = \sum_{j=1}^{\infty} \left\|T^* e_j'\right\|^2$$

by Parseval again. We also have

$$\sum_{i=1}^{\infty} \|T\hat{e}_i\|^2 = \sum_{j=1}^{\infty} \left\|T^* e_j'\right\|^2$$

Therefore, $\sum_{i=1}^{\infty} \|T\hat{e}_i\|^2 = \sum_{i=1}^{\infty} \|Te_i\|^2$ as required.

□

Proposition. Hilbert-Schmidt operators are compact.

Proof. Given an operator define the infinite matrix with coefficients $a_{ij} = \langle Te_j, e_i\rangle$.

Then $Tx = \sum_{i=1}^{\infty} \langle Tx, e_i\rangle e_i$ by an earlier lemma. Let $x^{(n)} = (x_1, x_2, ..., x_n, 0, ...)$ and note that $Tx^{(n)} = \sum_{i=1}^{\infty} \left(\sum_{j=1}^{n} a_{ij}x_j\right) e_i$. But note that $Tx^{(n)} \to Tx$ because T is continuousand that

$$\sum_{i=1}^{\infty} \left(\sum_{j=1}^{n} a_{ij}x_j\right) e_i \to \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij}x_j\right) e_i$$

Therefore $Tx = \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij}x_j\right) e_i$. If $x \in B_X(1)$ then we have $(Tx)_i = \sum_{j=1}^{\infty} a_{ij}x_j$ and by Cauchy-Schwarz we have

$$(Tx)_i \le \left(\sum_j |a_{ij}|^2\right)^{\frac{1}{2}} =: b_i$$

If T is Hilbert-Schmidt then

$$\sum_i |b_i|^2 = \sum_i\sum_j |a_{ij}|^2 = \sum_i\sum_j |\langle Te_j, e_i\rangle|^2 = \sum_j \|Te_j\|^2 < \infty$$

From this it follows that T is Hilbert-Schmidt if and only if $\sum_{i,j} |a_{ij}|^2 < \infty$.

It follows that $T(B_X(1))$ is contained in a Hilbert Cube given by

$$\{y = (y_1, y_2, ...) \in l_2 : |y_i| \le b_i\}$$

It remains to show that this Hilbert Cube is compact. Since we are in a metric space, it is enough to prove sequential compactness. We can do this by finding a subseqeuence $a_{1,n}$ that converges in the first term, a subsequence of that $a_{2,n}$ that converges in the second term and continuing to find these nested subseqeuences then noting that $\gamma_k = a_{k,k}$ is a pointwise convergent subsequence in the hilbert cube. Then this converges because for all $\varepsilon > 0$ there is an N such that $\sum_{n=N+1}^{\infty} |b_i|^2 < \frac{\varepsilon}{8}$ and then for k large enough, γ_k in the first N terms will always be within $\frac{\varepsilon}{2\sqrt{N}}$ of their limit, so we will only get a contribution of at most $\frac{\varepsilon}{2}$ from the first N terms and the tail so the subsequence converges in l_2.

□

Lemma. Let X, Y be Banach spaces, then $B(X, Y)$, the space of all bounded linear operators from X to Y, is a Banach space.

Proof. Take a Cauchy sequence (T_n) in $B(X,Y)$. Then for every $x \in X$ we have that the sequence $(T_n x)$ is Cauchy in Y. Since Y is complete $T_n x$ converges to a limit in Y and we will call this limit Tx, it is easy to check well definedness, linearity, etc. Therefore $B(X,Y)$ is complete.

□

Proposition. Let X, Y be Banach spaces and let $B_0(X,Y)$ be the set of all compact linear operators from X to Y. Then this is a closed linear subspace of $B(X,Y)$, the space of all bounded linear operators from X to Y.

Proof. A compact operator is bounded since $T(B_X(1))$ is bounded by compactness.

Given compact linear operators S, T we want to show that $S+T$ is compact. Note that it is clear that λT is compact for all $\lambda \in \mathbb{K}$.

To do this, let $(x_n)_{n=1}^{\infty}$ be a sequence of vectors with norm ≤ 1 in X. Then by compactness of S there exists a subsequence x'_n for which Sx'_n converges and a further subsequence x''_n for which Tx''_n also converges. Then $(S+T)x''_n$ converges.

Now to show closedness let $(T_n)_{n=1}^{\infty}$ be a sequence of compact operators with $T_n \to T$ in the operator norm. We must show that T is compact. Since this is equivalent to compactness of $T(B_X(1)) = \overline{T(B_X(1))}$ which is complete (closed subset of complete metric space), it is actually enough to show that $T(B_X(1))$ is totally bounded because totally bounded + complete is equivalent to compactness.

To do this, let $\varepsilon > 0$ and let n be such that $\|T_n - T\| < \frac{\varepsilon}{2}$. Since T_n is compact, $T_n(B_X(1))$ is totally bounded so there is a finite covering by $\frac{\varepsilon}{2}$-balls centered at $y_1, y_2, ..., y_m \in Y$. But then

$$\|Tx - y_i\| \leq \|Tx - T_n x\| + \|T_n x - y_i\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Since ε was arbitrary $T(B_X(1))$ is totally bounded.

□

Example. Let $(a_i)_{i=1}^{\infty}$ be a sequence of complex numbers tending to 0. Define $T : l_2 \to l_2$ by

$$T((x_1, x_2, ...)) = (a_1 x_1, a_2 x_2, ...)$$

Then this is bounded and linear and $\|T\| = \max_i |a_i|$.

Furthermore, T is compact as it is the limit in operator norm of the finite rank operators T_n defined by

$$T_n((x_1, x_2, ...)) = (a_1 x_1, a_2 x_2, ..., a_n x_n, 0, ...)$$

Definition. An operator is of finite rank if its image is finite dimensional. Observe that finite rank operators are automatically compact.

Proposition. Let X be a Hilbert space and let $T : X \to X$ be a compact operator. Then T is a limit of finite-rank operators.

Proof. Let $\varepsilon > 0$. Since $T(B_X(1))$ is totally bounded there are points $y_1, y_2, ..., y_m \in X$ such that for each $x \in B_X(1)$ there is i such that $\|Tx - y_i\| < \frac{\varepsilon}{2}$.

Let $Y = \langle y_1, y_2, ..., y_m \rangle$ be the linear span of these vectors and let $\pi : X \to Y$ be the orthogonal projection and set $\tilde{T} = \pi \circ T$ which has finite rank because Y is finite dimensional.

Now we will prove that $\left\|T - \tilde{T}\right\| < \varepsilon$. Let $x \in B_X(1)$ and choose y_i such that $\|Tx - y_i\| < \frac{\varepsilon}{2}$. Then

$$\left\|Tx - \tilde{T}x\right\| \leq \|Tx - y_i\| + \left\|y_i - \tilde{T}x\right\| \leq 2\|Tx - y_i\|$$

Note: The last inequality is because of how we defined $\tilde{T}$. Projecting onto Y like that decreases the distance to y_i.

Thus, for each $x \in B_X(1)$ we have $\left\| Tx - \tilde{T}x \right\| < \varepsilon$. Therefore, $\left\| T - \tilde{T} \right\| \leq \varepsilon$. Since ε was arbitrary, T is indeed a limit of finite-rank operators.

□

Proposition. Let X be a Banach space and suppose that $\lambda \neq 0$ is an approximate eigenvalue for some compact operator $T : X \to X$. Then λ is an eigenvalue of T.

Remark. Note that $\lambda \neq 0$ is necessary, since our earlier example of an approximate eigenvalue is a limit in operator norm of finite-rank truncations and therefore a compact operator, and yet it fails the above theorem for $\lambda = 0$. The reason we can't apply the above theorem to the 1 eigenvalue of $T + I$ is because that is not necessarily compact.

Proof. Let x_n be a sequence of unit vectors such that $(T - \lambda I) x_n \to 0$. Since $T(B_X(1))$ is compact, there is a convergent subsequence of the sequence Tx_n, say its limit is z and relabel the sequence to this subsequence. But then $z - \lambda x_n = (z - Tx_n) + (Tx_n - \lambda x_n) \to 0$. Therefore, $x_n \to \frac{1}{\lambda} z =: z^*$, but then z^* is a (unit) eigenvector of T with eigenvalue λ because

$$(T - \lambda I) z^* = \lim_{n \to \infty} (T - \lambda I) x_n = 0$$

□

7.3 The spectral theorem for compact, self-adjoint operators on a Hilbert space

Recall from Linear algebra that self adjoint matrices have real eigenvalues with an orthonormal basis of eigenvectors. We aim to prove the analogous result in a Hilbert space. The Spectral theorem stated below is the main result of this subsection.

Theorem. (Spectral Theorem for compact, self-adjoint operators on a Hilbert space) Let X be a Hilbert space and let $T : X \to X$ be a compact self-adjoint operator. Then

1. All eigenvalues are real

2. Eigenvectors corresponding to distinct eigenvalues are orthogonal

3. All Eigenspaces are finite dimensional except possibly the zero space

4. There are only finitely many eigenvalues λ with $|\lambda| \geq \varepsilon$ for each $\varepsilon > 0$

5. Every element of X can be written as the sum of its projections onto the eigenspaces of every eigenvector.

6. Any operator satisfying properties 1-5 is compact and self-adjoint.

We will prove this once we prove a few preliminary lemmas.

Lemma. In this subsection, from now on X is a hilbert space. Let $T : X \to X$ be self-adjoint, then $(\operatorname{im}(T))^{\perp} = \ker(T)$.

Proof. If $x \in (\operatorname{im}(T))^{\perp}$ then $\langle x, Ty \rangle = 0$ for all y and therefore $\langle Tx, y \rangle = 0$ for all y, so $Tx = 0$. If $x \in \ker(T)$ then the identical proof in reverse proves $x \in (\operatorname{im}(T))^{\perp}$.

□

Lemma. Let $T : X \to X$ be self-adjoint. Then if $\lambda \in \sigma(T)$ then λ is an approximate eigenvalue for T

Proof. We saw before that if $\operatorname{im}(T - \lambda I)$ is dense in X then λ is an approximate eigenvalue. Therefore it suffices to show that $\operatorname{im}(T - \lambda I)$ is dense in X. Without loss of generality assume $\lambda = 0$.

If 0 is an approximate eigenvalue then we're done. Therefore suppose then that 0 is not an approximate eigenvalue. Then T is bounded below, say $\|Tx\| \geq \varepsilon \|x\|$. But $(\operatorname{im}(T))^{\perp} = \ker(T)$ which is 0 because T is bounded below. Thus, $(\operatorname{im}(T))^{\perp} = 0$ so $\overline{\operatorname{im}(T)} = (\operatorname{im}(T))^{\perp\perp} = 0^{\perp} = X$. Therefore $\operatorname{im}(T)$ is dense in X.

□

Remark. We noted earlier that approximate eigenvalues are always in $\sigma(T)$ so the set of approximate eigenvalues is equal to $\sigma(T)$, provided the operator is self-adjoint.

Corollary. If T is compact and self-adjoint then $\sigma(T)$ consists of all approximate eigenvalues of T and therefore by a previous proposition $\sigma(T) \setminus \{0\}$ consists of all eigenvalues.

Lemma. Let T be compact and self-adjoint. Then T has at least one real eigenvalue (in fact it is either $\|T\|$ or $-\|T\|$).

Proof. The result is trivial if $T = 0$ so assume not.

We will show that $T^2 - \|T\|^2 I$ is not invertible, which implies by factoring it that at least one of $T - \|T\| I$ and $T + \|T\| I$ is not invertible, and thus by the above corollary the result follows.

It is enough to find an approximate eigenvector by compactness and an earlier theorem, and the fact that the set of approximate eigenvalues coincides with the spectrum.

Take a sequence (x_n) of unit vectors such that $\|Tx_n\| \to \|T\|$. Then we have

$$\left\|T^2 x_n - \|T\|^2 x_n\right\|^2 = \left\|T^2 x_n\right\|^2 + \|T\|^4 - \|T\|^2 \left(\left\langle x_n, T^2 x_n\right\rangle + \left\langle T^2 x_n, x_n\right\rangle\right)$$

$$= \left\|T^2 x_n\right\|^2 + \|T\|^4 - 2\|T\|^2 \|Tx_n\|^2$$

by self-adjoint-ness.

$$\leq \left\|T^2\right\|^2 \|x_n\|^2 + \|T\|^4 - 2\|T\|^2 \|Tx_n\|^2$$

$$\leq 2\|T\|^4 - 2\|T\|^2 \|Tx_n\|^2 \to 0$$

by definition of x_n.

Thus, $\left(T^2 - \|T\|^2 I\right) x_n \to 0$ so it has an approximate eigenvalue, therefore it is not invertible

□

We will now prove the spectral theorem.

Proof. **1,2.** That all eigenvalues are real is true for the same reason as the finite dimensional case. That is, we get $\lambda = \bar{\lambda}$ because

$$\lambda \|x\|^2 = \langle Tx, x\rangle = \langle x, Tx\rangle = \bar{\lambda} \|x\|^2$$

and $\|x\|^2 \neq 0$. The proof that eigenvectors with distinct eigenvalues are orthogonal is similar, and it is also essentially the same as the finite dimensional case, details are left as an exercise.

3,4. Note that if either of parts 3 or 4 of the theorem fail, then there exists an infinite orthonormal sequence $(x_n)_{n=1}^{\infty}$ with $Tx_n = \lambda_n x_n$ with $|\lambda_n| \geq \varepsilon$ for all n and $\varepsilon > 0$. (Note: if part 3 fails then use Gram-Schmidt on a countable dimension eigenspace and if part 4 fails pick one unit vector from each eigenspace). But then

$$\|Tx_n - Tx_m\|^2 = \|\lambda_n x_n - \lambda_m x_m\|^2 = \lambda_n^2 + \lambda_m^2 \geq 2\varepsilon^2$$

Therefore this does not have a convergent subsequence contradicting compactness.

Note that there are countably many eigenvalues, because they are a countable union of finite sets: Specifically,

$$\sigma(T) = \{\text{possibly } 0\} \cup \bigcup_{n=1}^{\infty} \left(\sigma(T) \cap \left(\mathbb{R} \setminus \left(-\frac{1}{n}, \frac{1}{n} \right) \right) \right)$$

5. To prove part 5 of the theorem, let V be the closed linear span of the eigenspaces, and write $X = V \oplus V^{\perp}$.

Now let $x \perp E_{\lambda}$. Then $\langle x, y \rangle = 0$ for all $y \in E_{\lambda}$ so

$$\langle Tx, y \rangle = \langle x, Ty \rangle = \lambda \langle x, y \rangle = 0$$

Therefore $Tx \perp E_{\lambda}$.

By applying this simaltaneously to all eigenspaces it follows that $Tx \in V^{\perp}$ if $x \in V^{\perp}$. It follows that T restricted to $V^{\perp}$ is a genuine map, and that it has no non-zero eigenvalues, therefore it is 0.

Now pick an orthonormal basis for each eigenspace together with any orthonormal basis for $V^{\perp}$ (Note: this exists by the same Zorn's lemma argument that proved every vector space has a basis). Then this gives an orthonormal basis consisting of the eigenvectors of T.

Now suppose this basis consists of eigenvectors $e_i : i \in I$ with $Te_i = \lambda_i e_i$ together with a basis $f_j : j \in J$ for $V^{\perp}$. Then

$$T \left(x_1 e_1 + x_2 e_2 + ... + \text{something in } V^{\perp} \right)$$

$$= \lambda_1 x_1 + \lambda_2 x_2 + ... + 0$$

since T is bounded, it is continuous, thus limits are indeed preserved when we take the infinite sum like that.

6. We claim that if $T = \sum_{n=1}^{\infty} \lambda_n P_n$ where P_n are mutually orthogonal projections onto finite dimensional spaces and $\lambda_n \to 0$ and they are real, then T is compact and self-adjoint.

Each P_n is self-adjoint because projections onto complete subspaces are self-adjoint. Therefore any finite linear combination of P_n is also self-adjoint. Since self-adjoint-ness is preserved by limits in the operator norm, T is also self-adjoint.

Also, each P_n is a finite-rank operator, thus compact, so the partial sums $T_N = \sum_{n=1}^{N} \lambda_n P_n$ are finite linear combinations of compact operators and thus also compact. Since the tail $\|T - T_N\|$ tends to 0 as $\sup_{n>N} |\lambda_n|$ tends to 0, T is the norm limit of compact finite rank operators, thus also compact by an earlier theorem.

$\square$